

Lecture#5: Representation Learning on Hyper-relational Knowledge Graphs

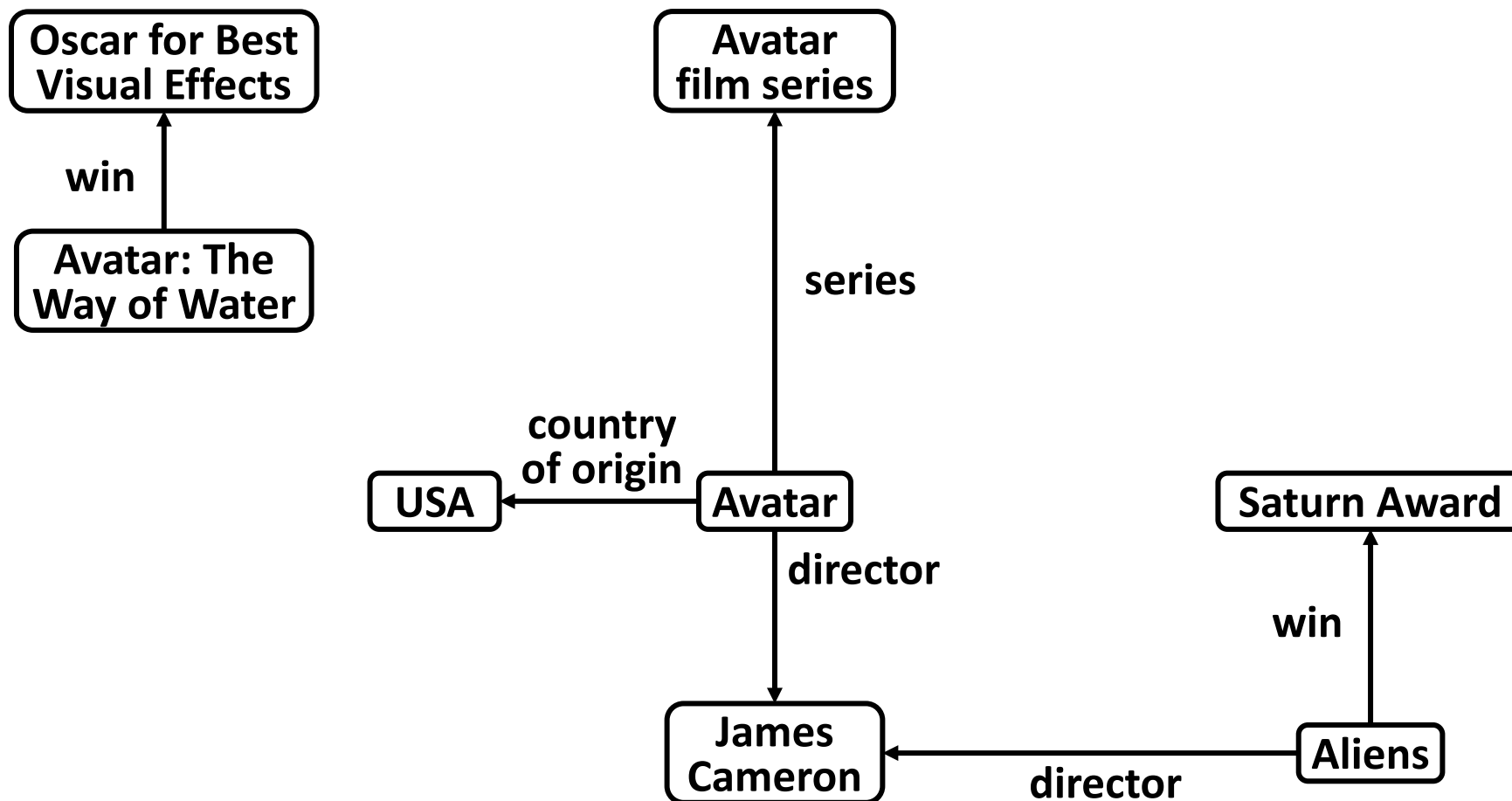
Jaejun Lee

School of Computing, KAIST

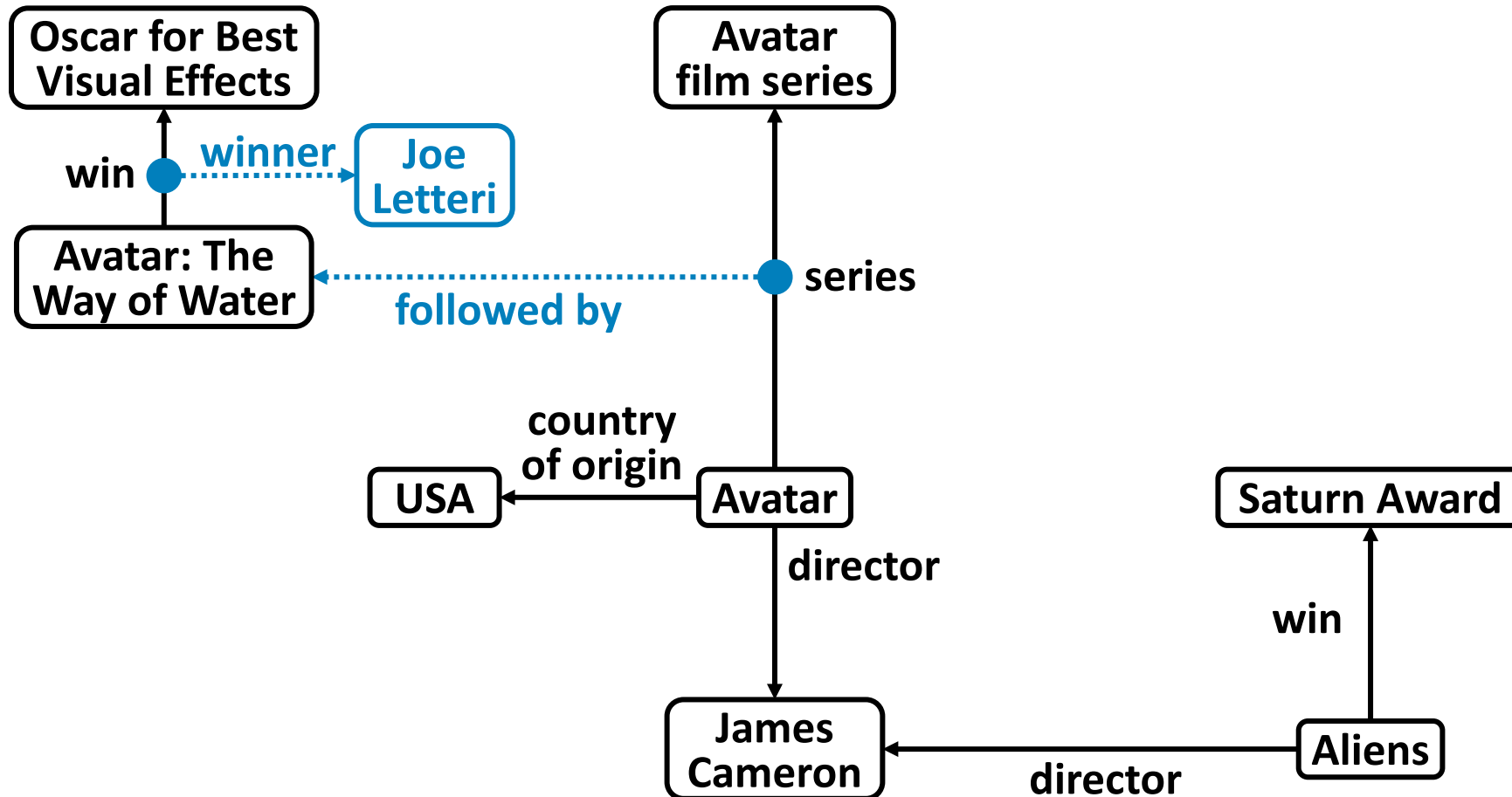
Knowledge Discovery with Knowledge Graphs: From Structural Embeddings to Generative Reasoning (KnowKG), KDD 2026 Tutorial

<https://bdi-lab.kaist.ac.kr>





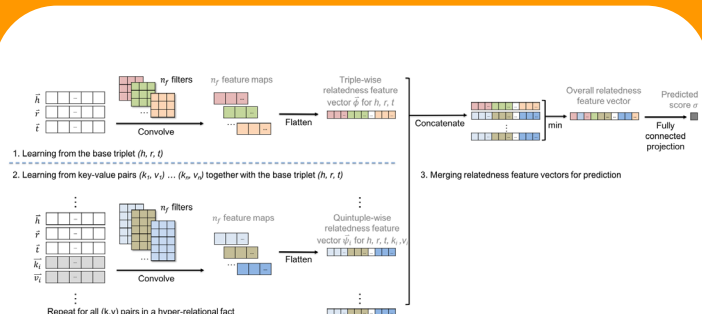
01 Hyper-relational Knowledge Graphs (HKGs)



01 Representation Learning on HKGs

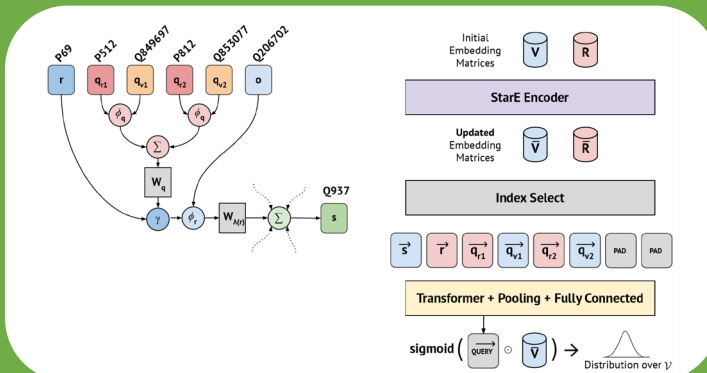
HINGE (TheWebConf 2020)

- A hyper-relational KG embedding model that directly learns from hyper-relational facts in a KG
- Captures the structural information of the KG encoded in the triplets
 - Also captures the correlation between each triplet and its associated qualifiers



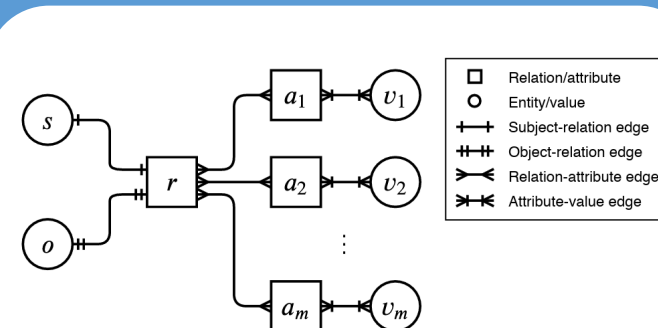
StarE (EMNLP 2020)

- A message passing based graph encoder that is capable of modeling hyper-relational KGs
- Can encode qualifiers along with the main triplet
- Demonstrates that existing benchmarks for evaluating LP on HKGs suffer from flaws



GRAN (ACL 2021 Findings)

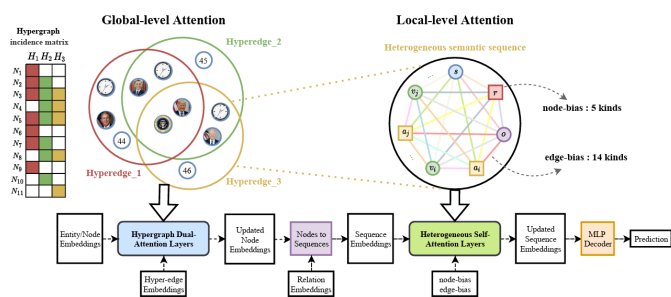
- Represents the structure of a hyper-relational fact as a small heterogeneous graph
 - Models the heterogeneous graph with edge-biased fully-connected attention
- Can fully model global and local dependencies in each fact



01 Representation Learning on HKGs

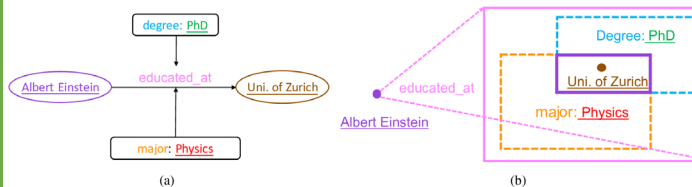
HAHE (ACL 2023)

- Represents the global structure of HKG as a hypergraph and the local structure as a semantic sequence
- Separately models the graphical structure of HKGs and the sequential structure inside facts
- Performs multi-position prediction in hyper-relational KGs



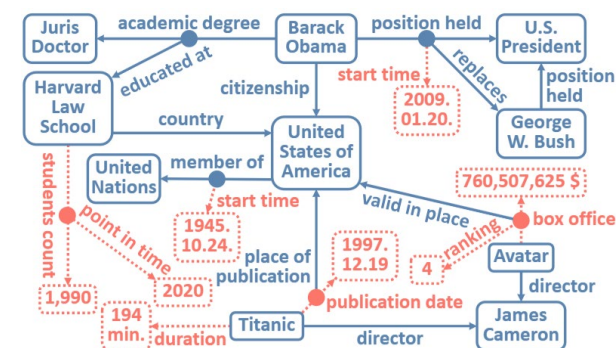
ShrinkE (ACL 2023)

- A geometric HKG embedding method aiming to explicitly model essential inference patterns of facts
- Models a primary triplet as a spatial-functional transformation from the head into a relation-specific box
 - Each qualifier shrinks the box to narrow down the answer set



HyNT (KDD 2023)

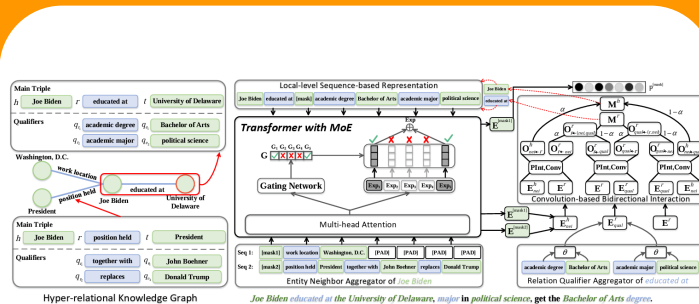
- A unified framework that learns representations of a HKG containing numeric literals in triplets/qualifiers
- Reduces the cost of transformers by learning compact representations of triplets and qualifiers
- Predicts missing entities, relations, and numeric values in a HKG



01 Representation Learning on HKGs

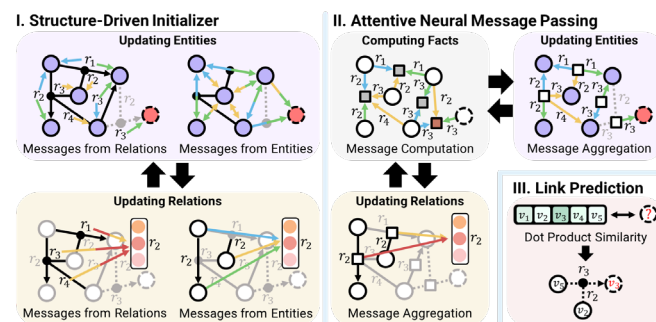
HyperFormer (CIKM 2023)

- A model that considers local-level sequential information
 - Encodes the content of the entities, relations, and qualifiers of a triplet
- Introduces a Mixture-of-Experts strategy to strengthen the representation capabilities



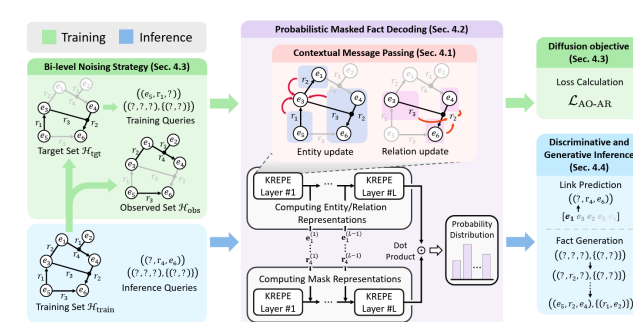
MAYPL (ICML 2025)

- Demonstrates that thoroughly leveraging the structure of an HKG is crucial for reasoning on HKGs
- The first structural representation learning method for HKGs that can be applied in both transductive and inductive learning settings



KREPE (ICML 2026)

- Proposes fact generation, a task of generating a valid fact from an arbitrarily masked query
- The first generative representation learning method for HKGs
 - models the joint conditional probability distributions of missing components in a query

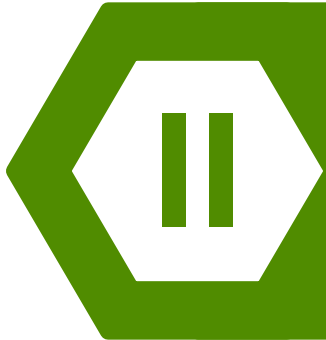
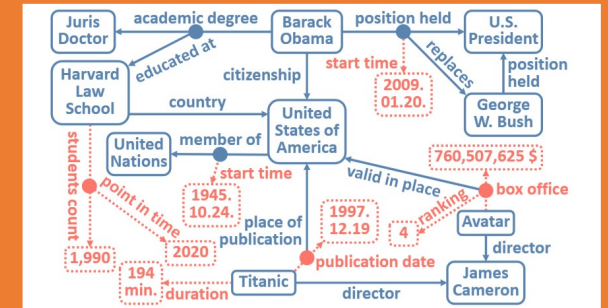




Representation Learning on Hyper-Relational and Numeric Knowledge Graphs with Transformers

Chanyoung Chung[‡], Jaejun Lee[‡], and Joyce Jiyoung Whang^{*}

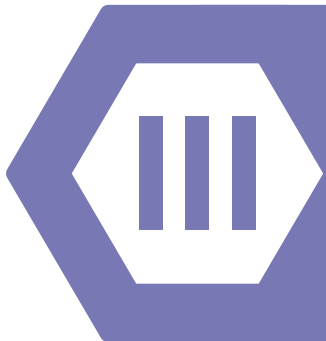
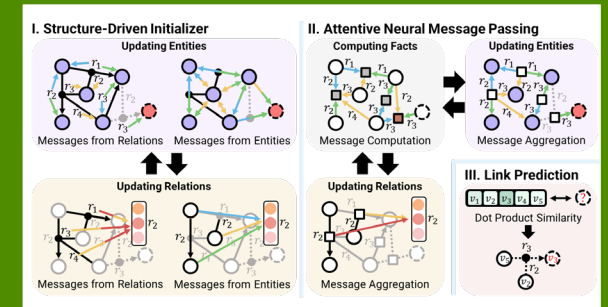
KDD 2023



Structure Is All You Need: Structural Representation Learning on Hyper-Relational Knowledge Graphs

Jaejun Lee and Joyce Jiyoung Whang^{*}

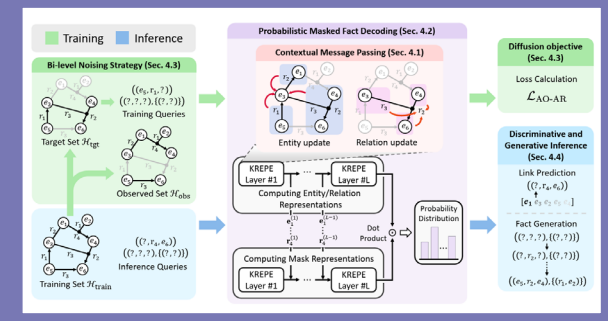
ICML 2025



Generative Representation Learning on Hyper-relational Knowledge Graphs via Masked Discrete Diffusion

Jaejun Lee, Seheon Kim, and Joyce Jiyoung Whang^{*}

ICML 2026

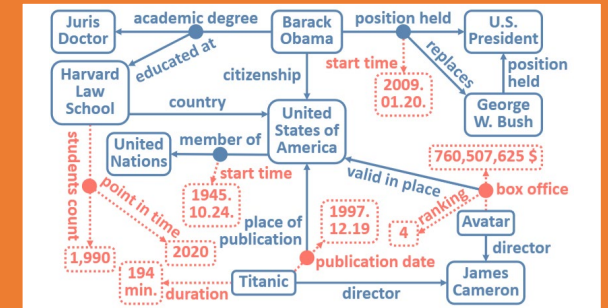




Representation Learning on Hyper-Relational and Numeric Knowledge Graphs with Transformers

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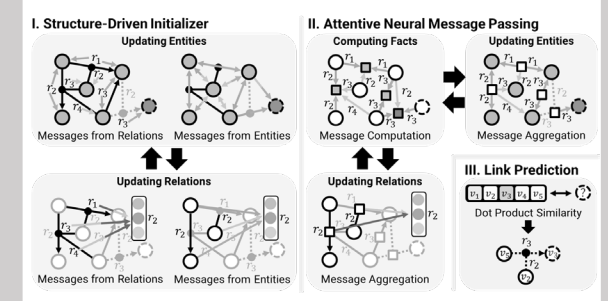
KDD 2023



Structure Is All You Need: Structural Representation Learning on Hyper-Relational Knowledge Graphs

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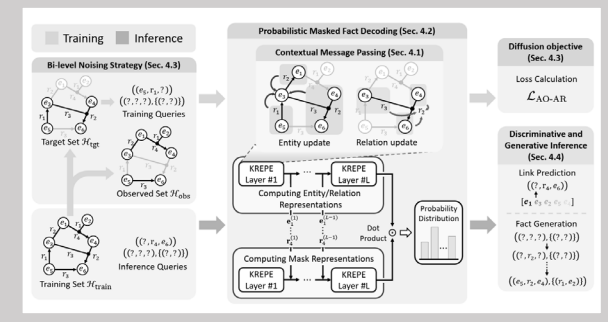
ICML 2025



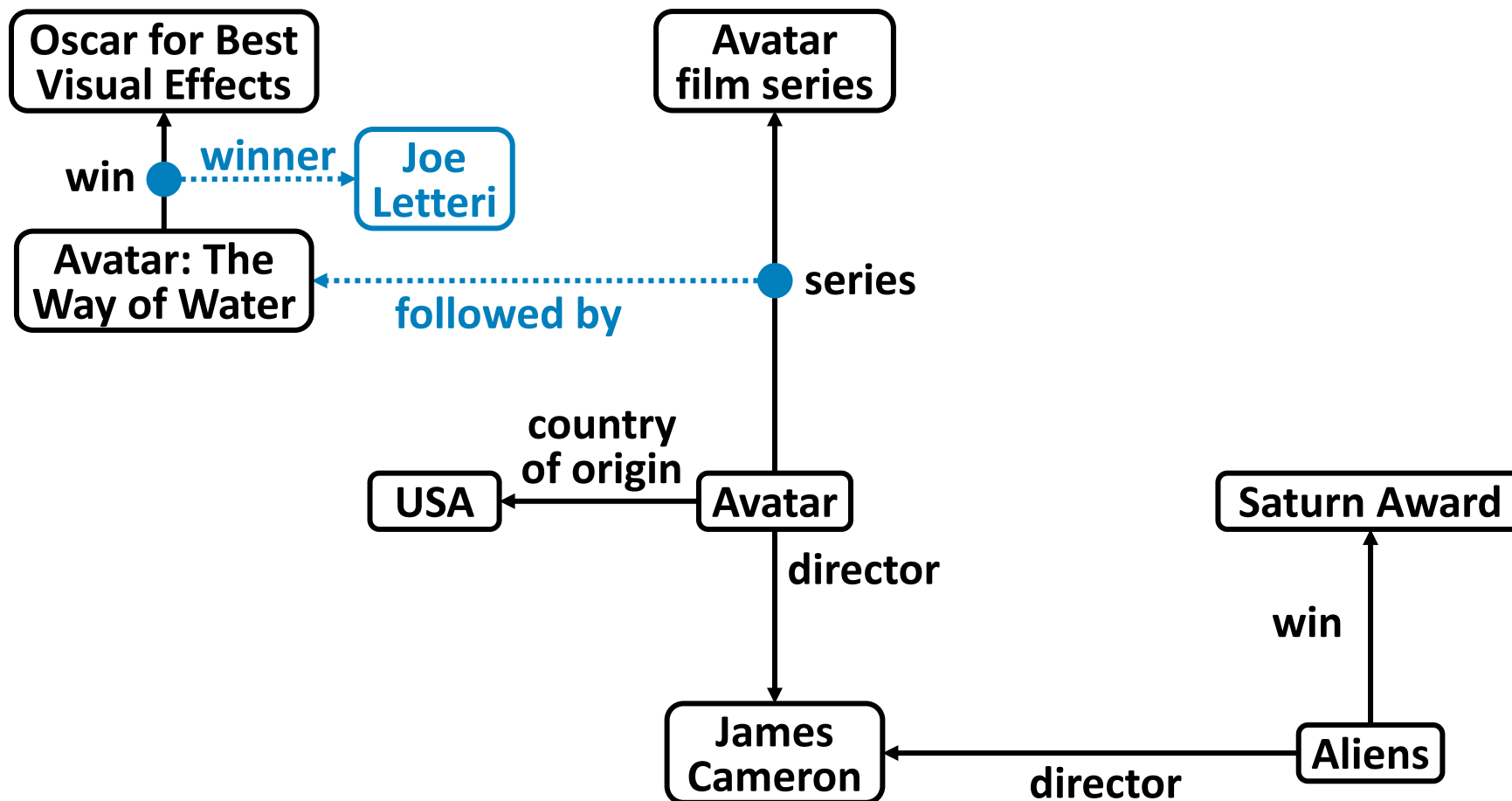
Generative Representation Learning on Hyper-relational Knowledge Graphs via Masked Discrete Diffusion

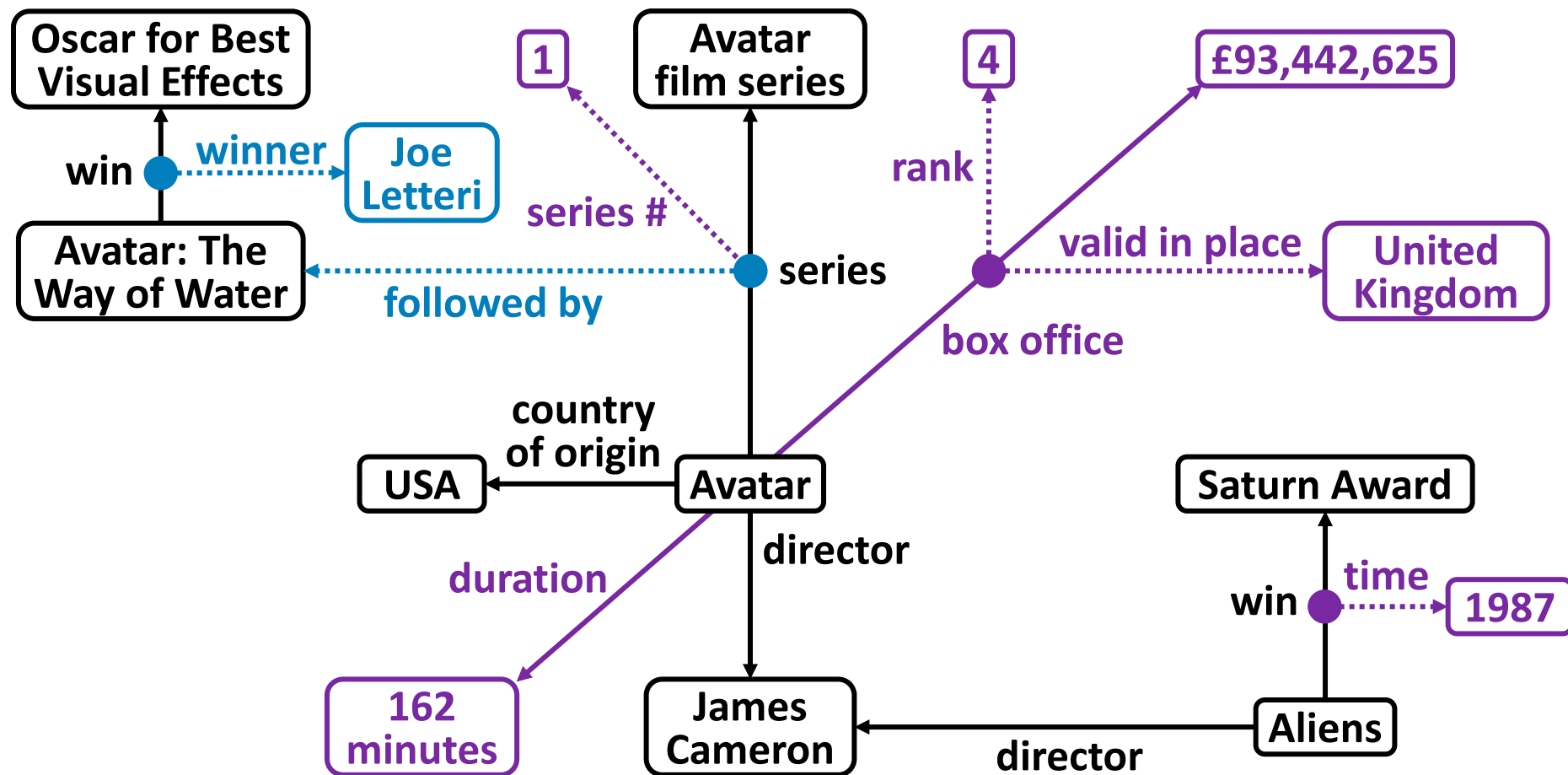
Jaejun Lee, Seheon Kim, and Joyce Jiyoung Whang^{*}

ICML 2026



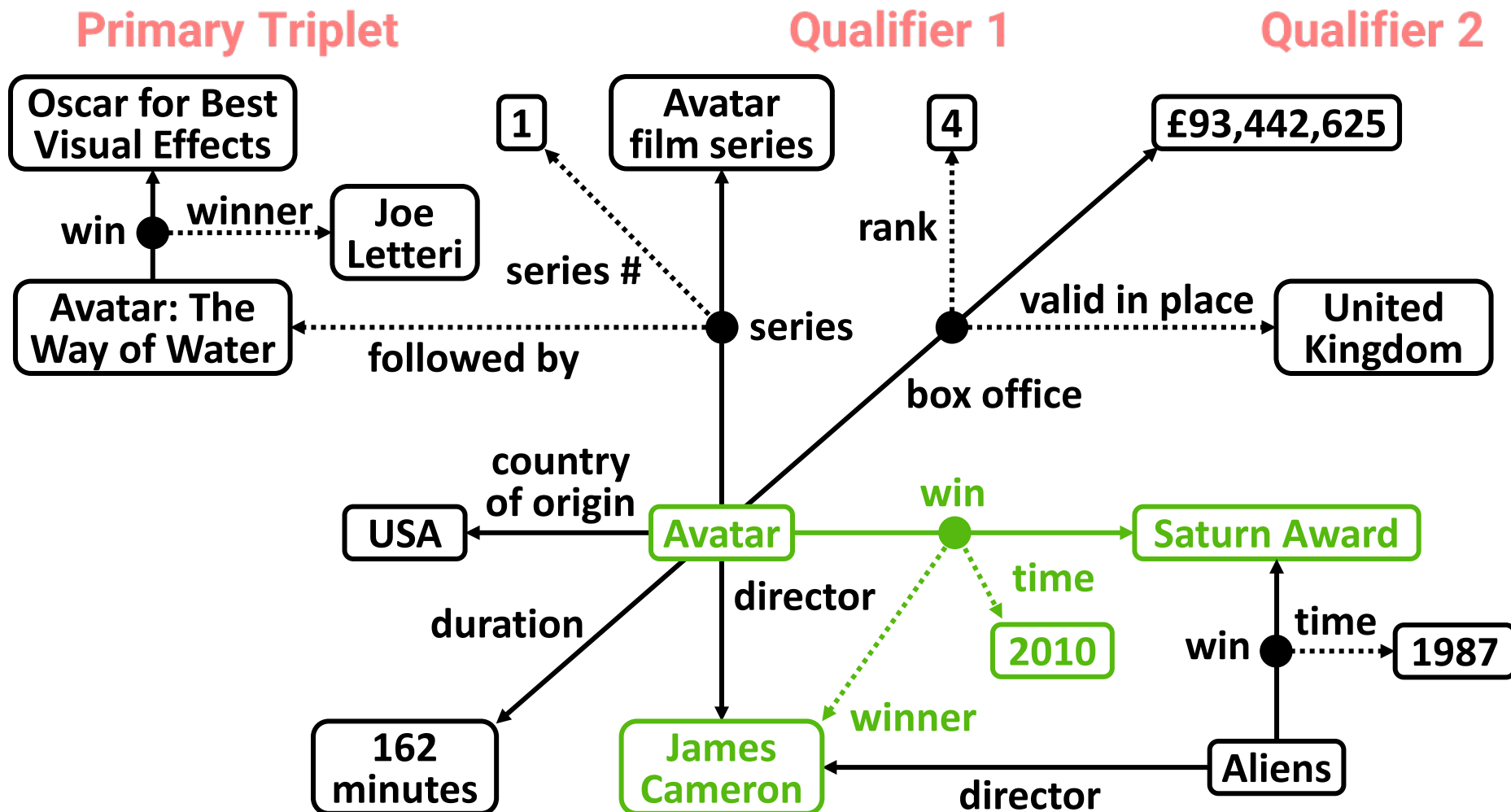
Hyper-relational Knowledge Graphs





02 Hyper-relational and Numeric Knowledge Graphs

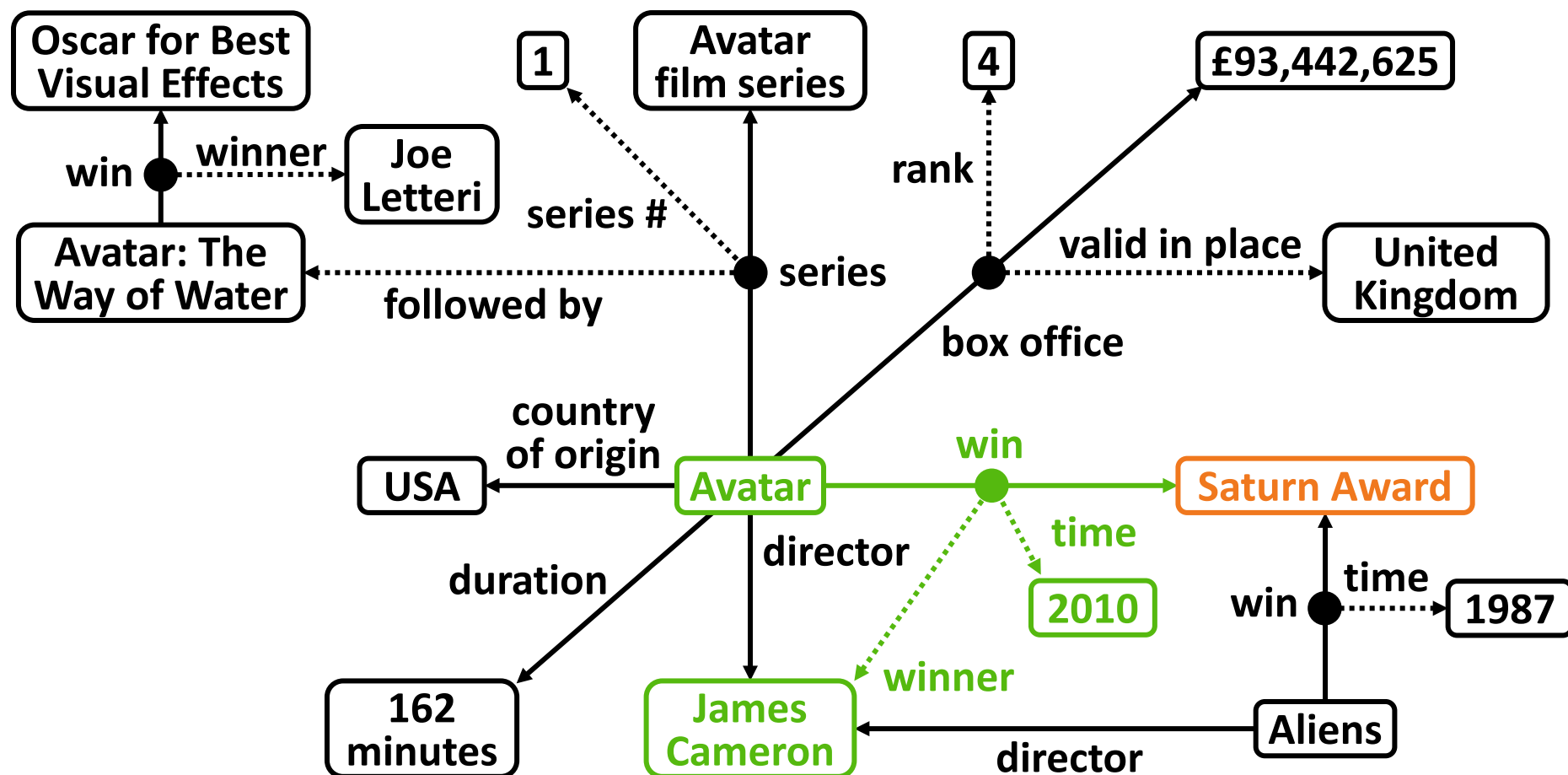
((Avatar, win, Saturn_Award), {(winner, James_Cameron), (time, 2010)}))



02

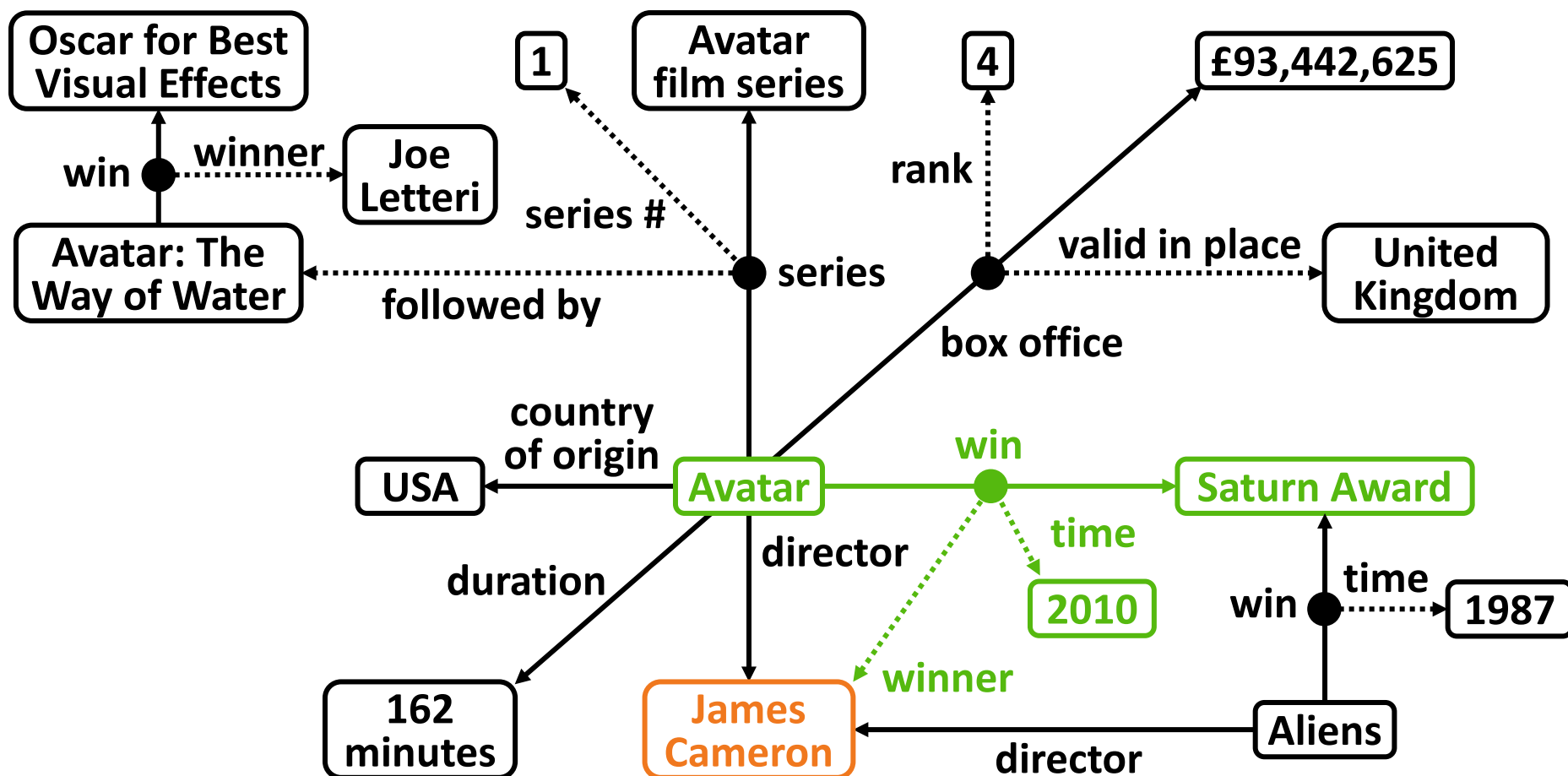
Link Prediction on HN-KGs

((Avatar, win, ?), {(winner, James_Cameron), (time, 2010)})



02 Link Prediction on HN-KGs

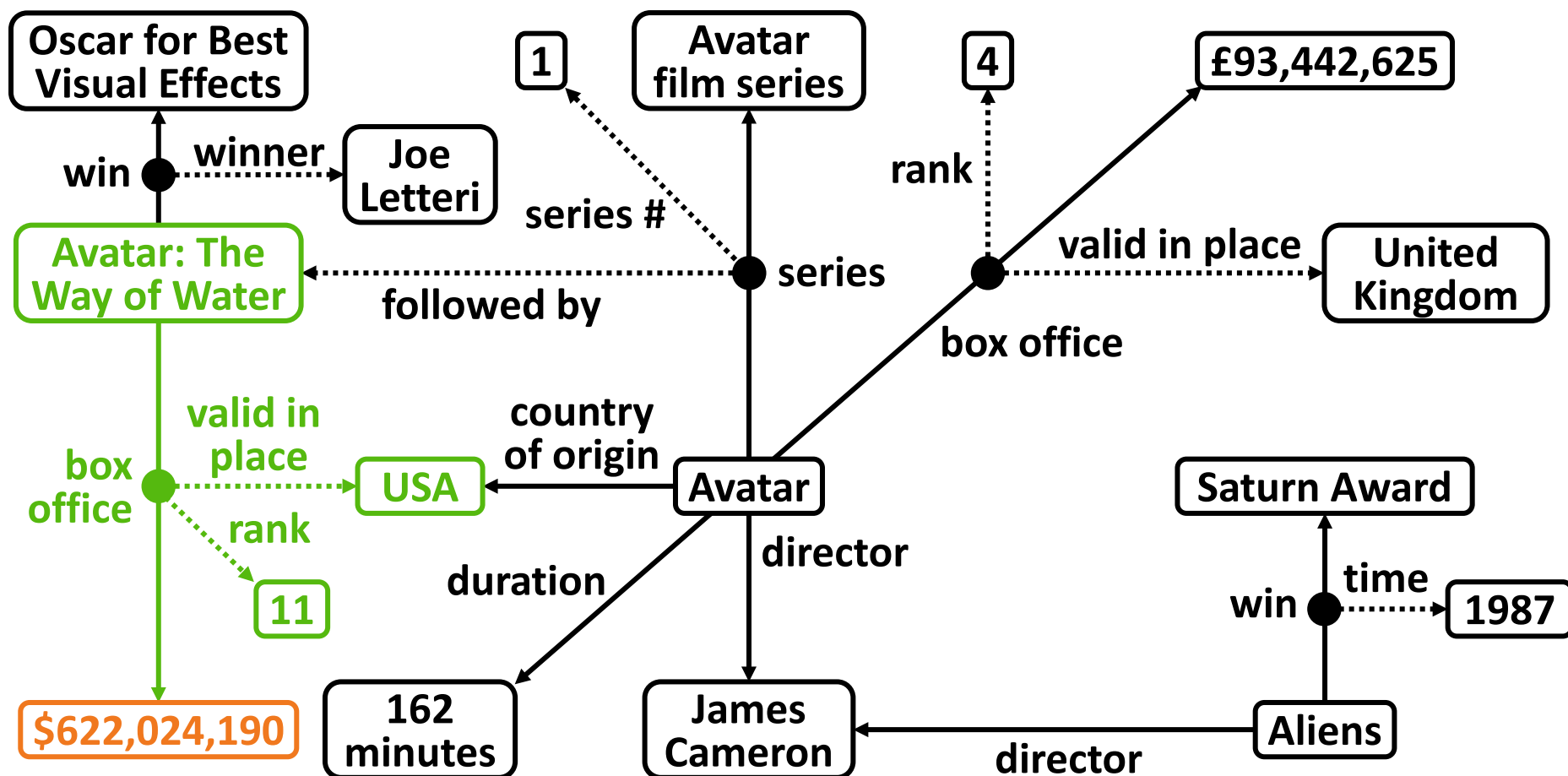
$((\text{Avatar}, \text{win}, \text{Saturn_Award}), \{(\text{winner}, \text{James_Cameron}), (\text{time}, 2010)\})$



02

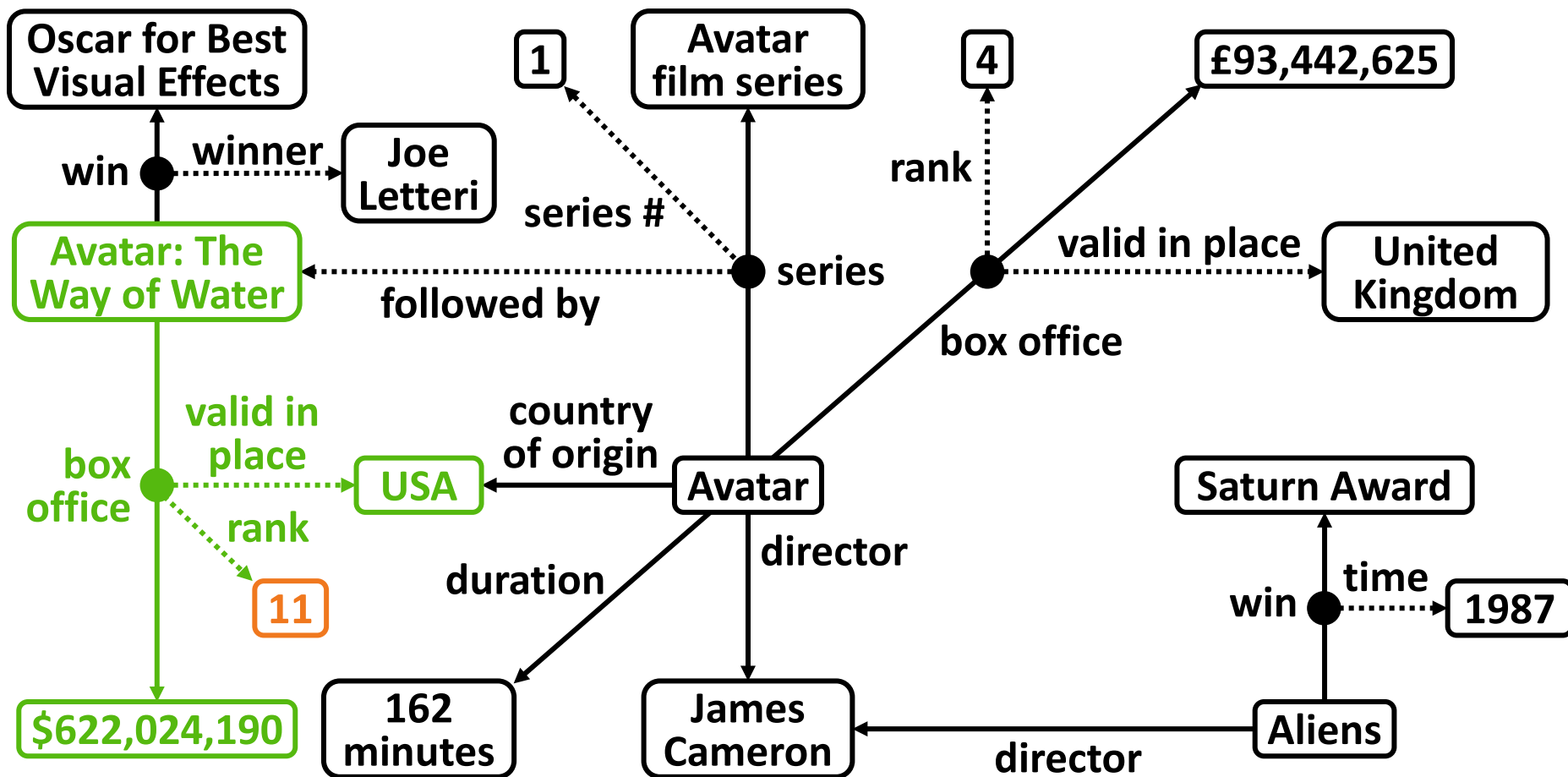
Numeric Value Prediction on HN-KGs

((Avatar:The_Way_of_Water, box_office, ?), {(rank, 11), (valid_in_place, USA)}))



Numeric Value Prediction on HN-KGs

((Avatar:The_Way_of_Water, box_office, \$622,024,190), {(rank, ?), (valid_in_place, USA)})

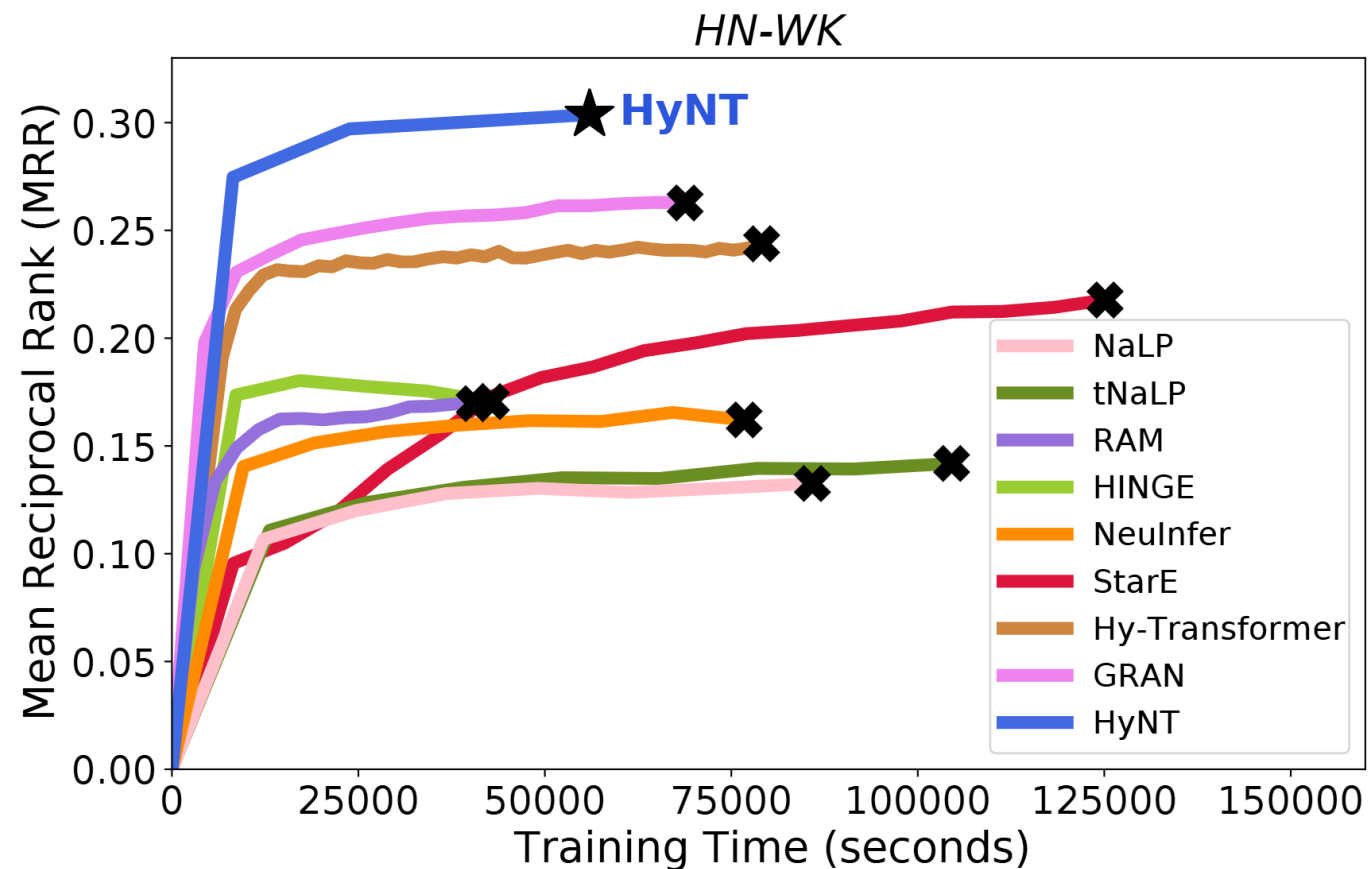


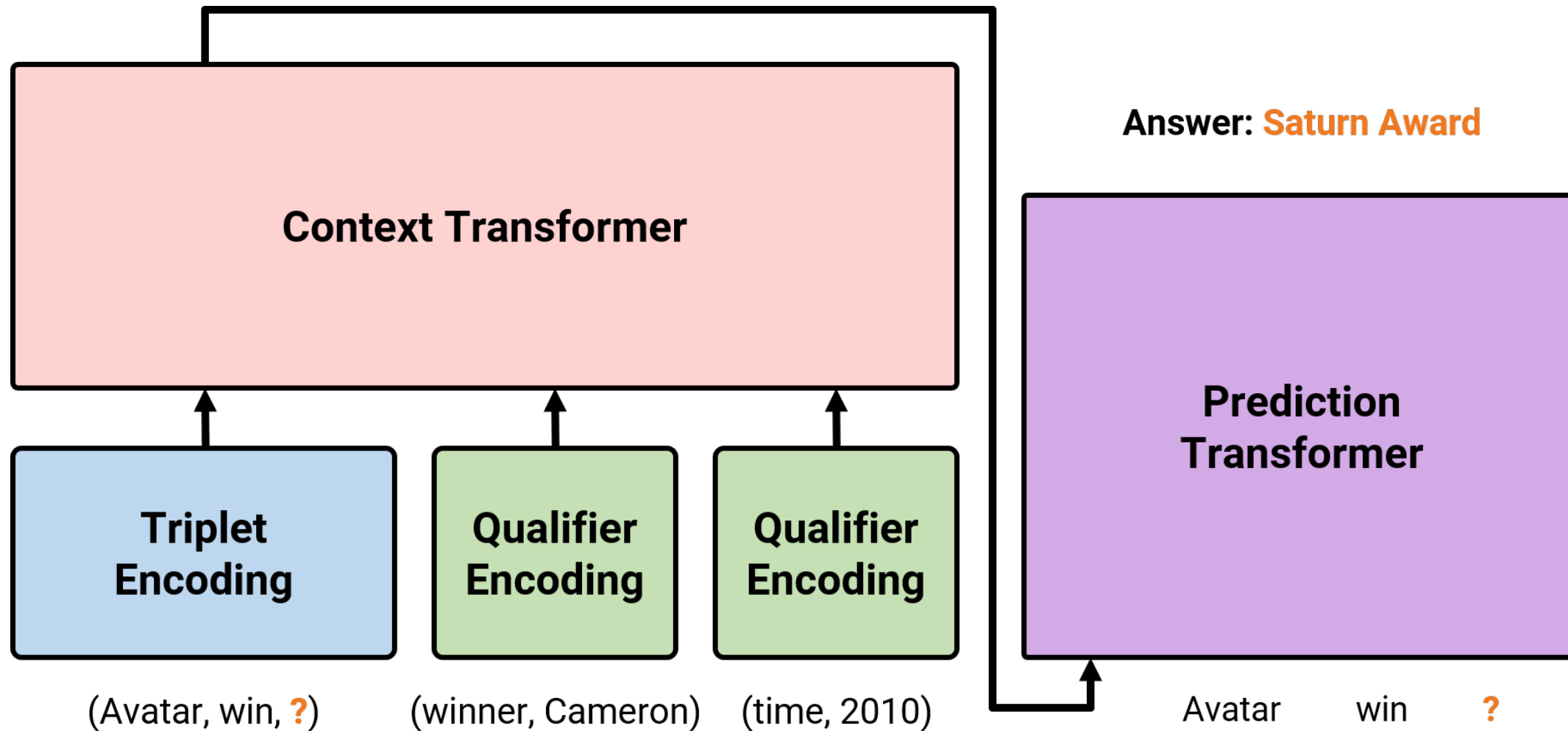
02 Contributions

- Define **Hyper-relational and Numeric Knowledge Graphs**
 - Create 4 real-world HN-KG datasets
- Propose **HyNT**, Hyper-relational knowledge graph embedding with **N**umeric literals using **T**ransformers
 - Define a context transformer and a prediction transformer
 - Reduce the cost by learning compact representations of triplets and qualifiers
- HyNT significantly outperforms 12 different state-of-the-art methods for **link prediction**, **numeric value prediction**, and **relation prediction**

02 Contributions

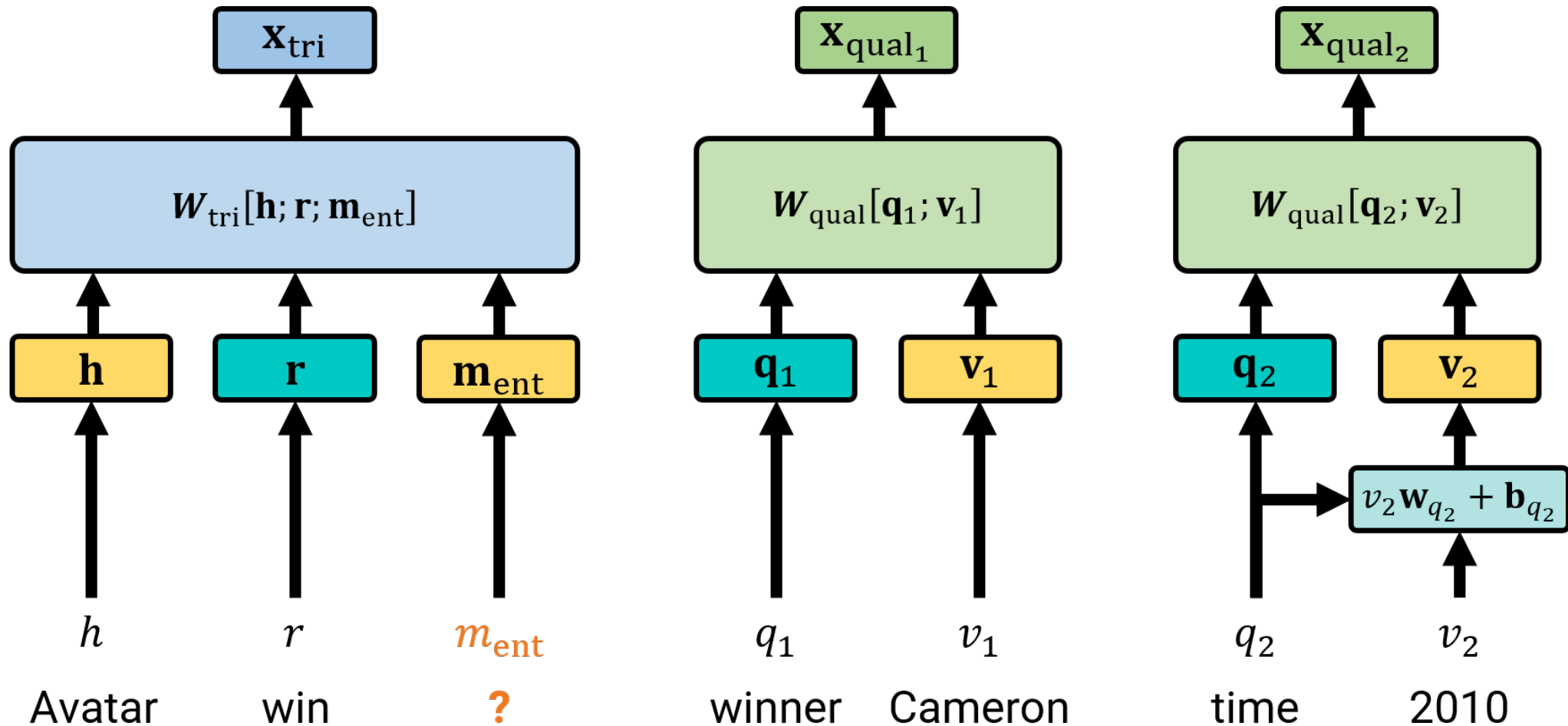
- Link Prediction Performance vs. Training Time



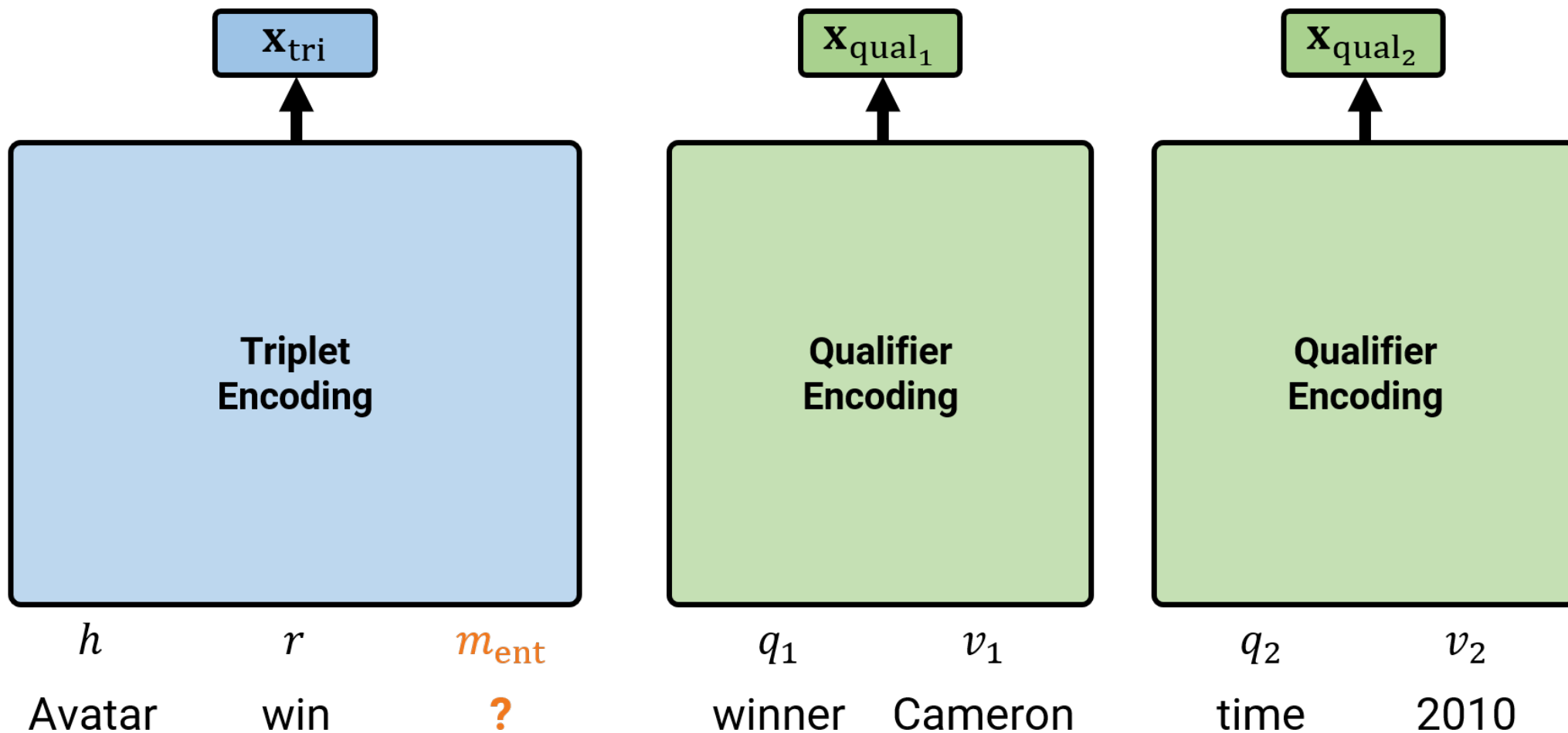


02

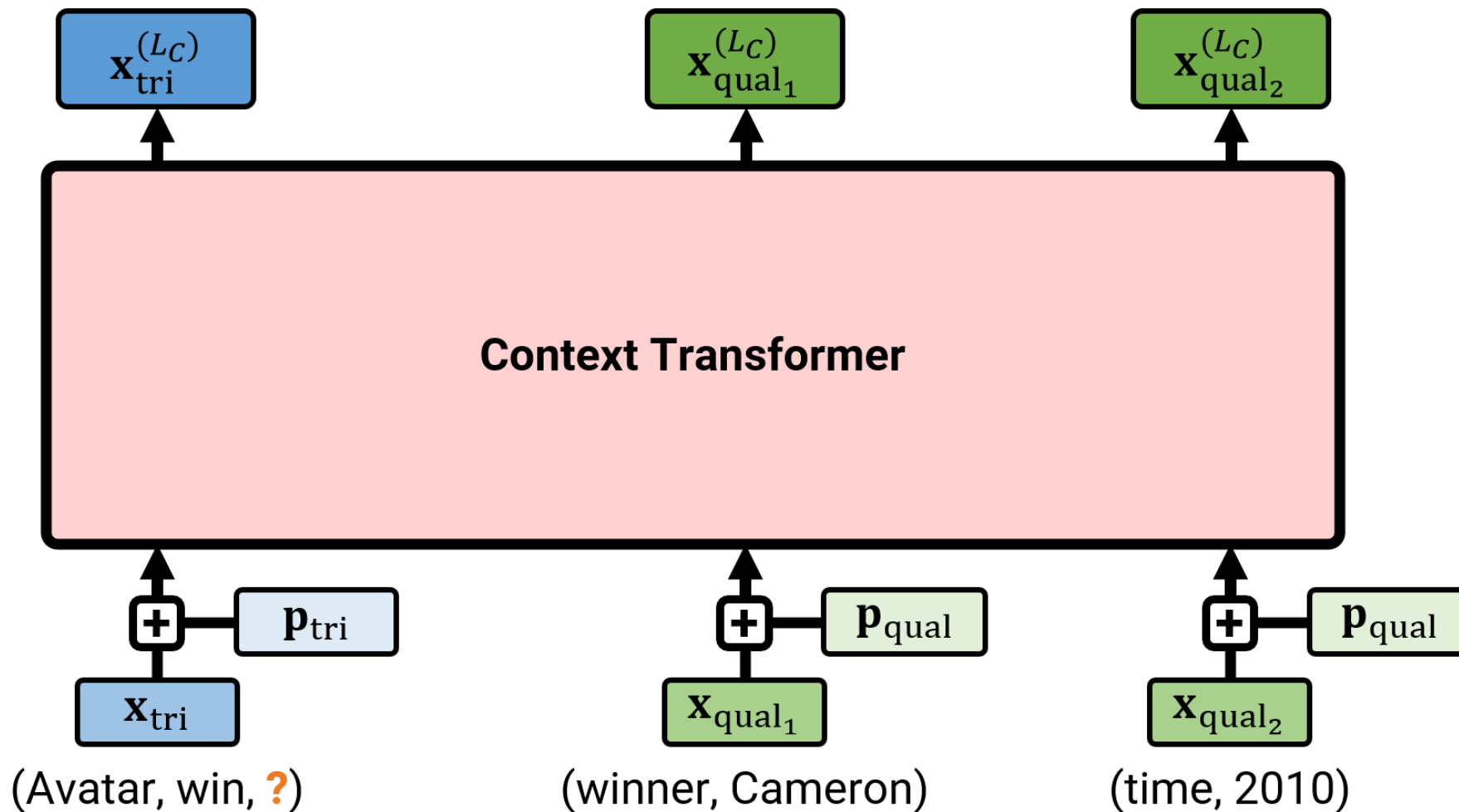
Triplet/Qualifier Encoding



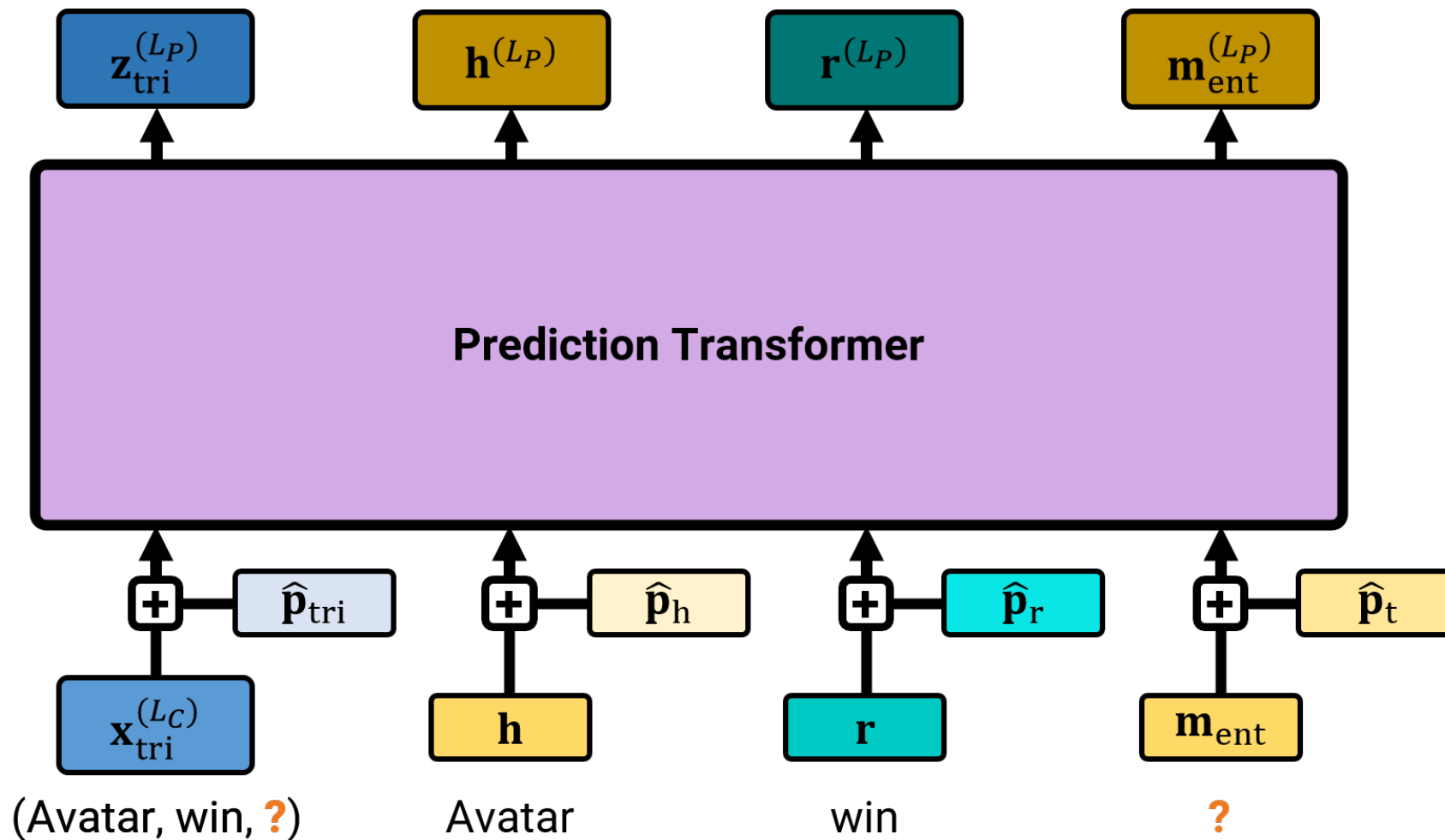
Triplet/Qualifier Encoding



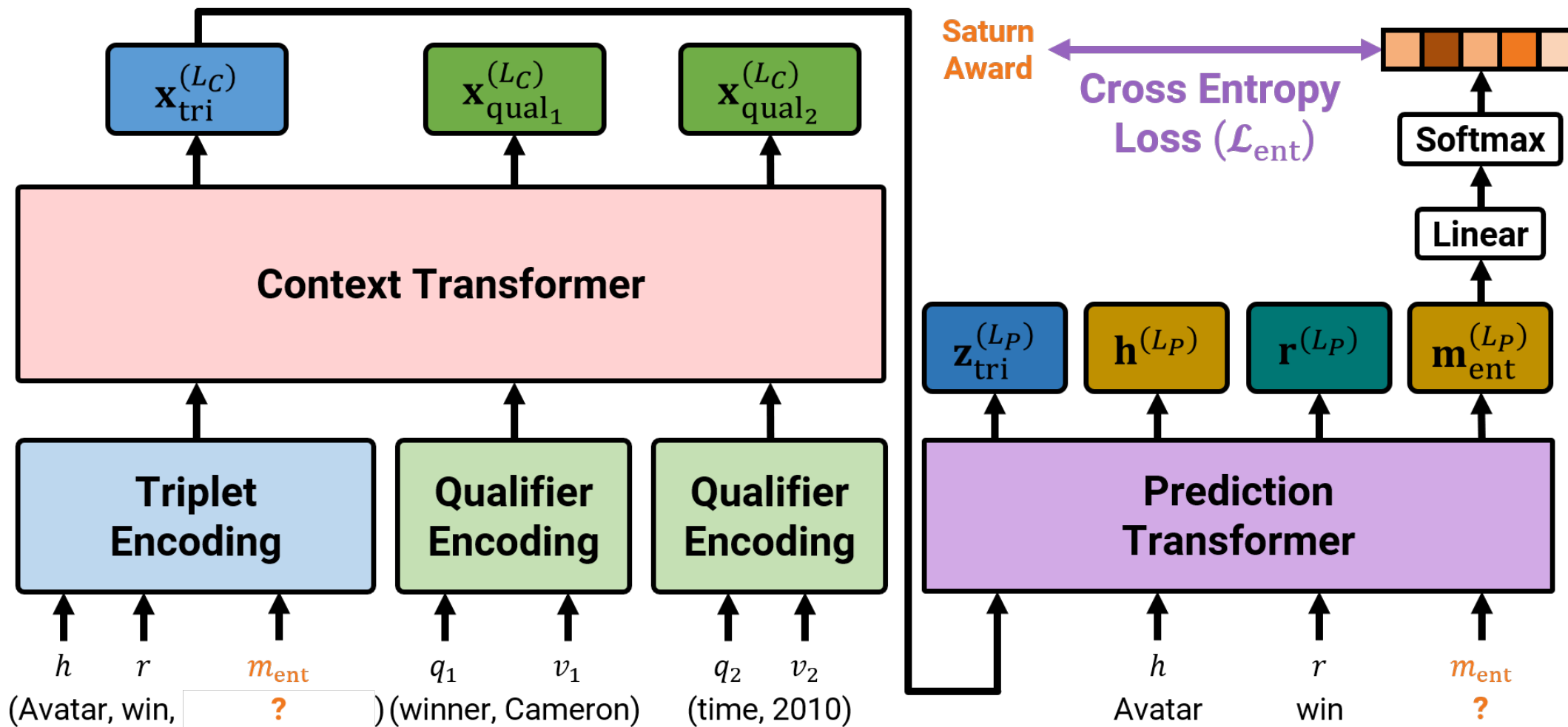
Context Transformer



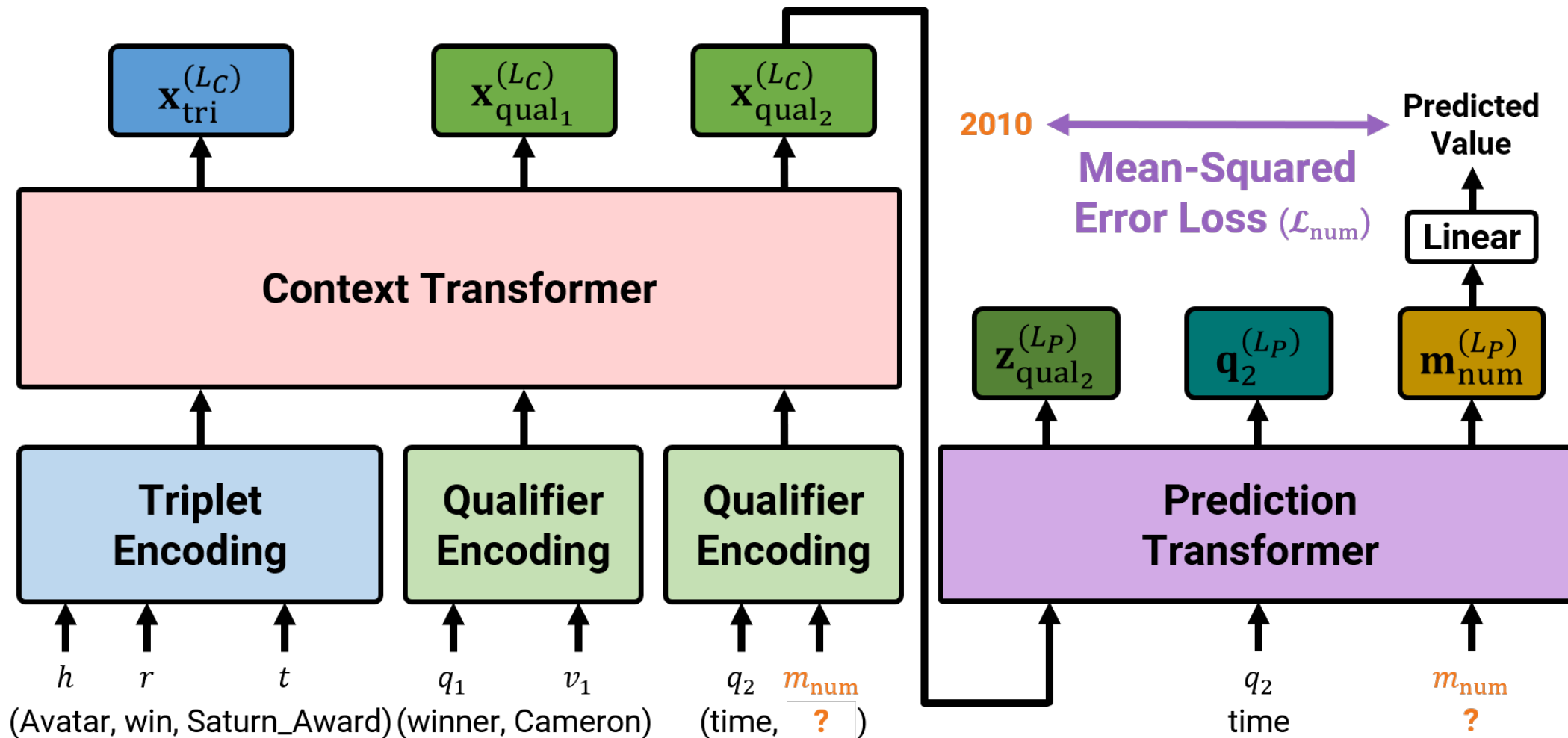
Prediction Transformer



Link Prediction using HyNT



Numeric Value Prediction using HyNT

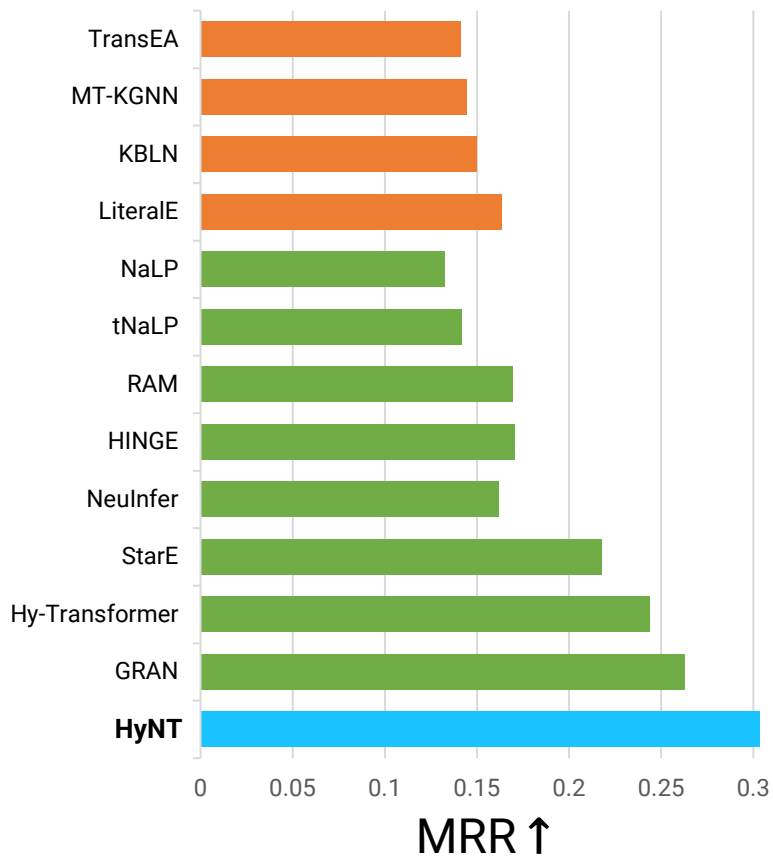


02 Experimental Results

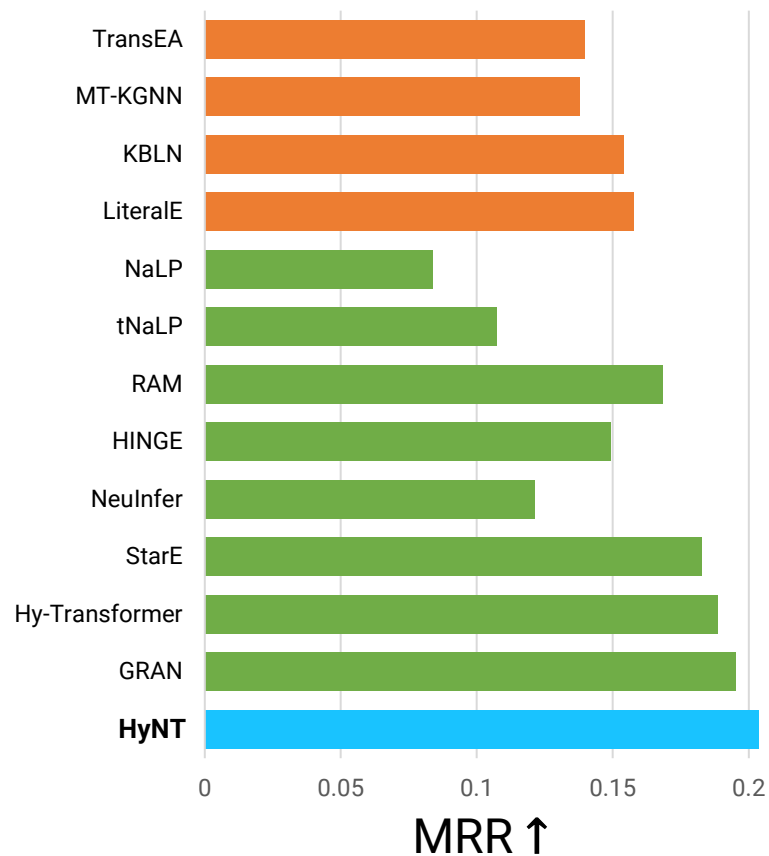
- Datasets
 - Based on Wikidata, YAGO, and Freebase
 - Create **4 Hyper-relational and Numeric Knowledge Graph (HN-KG)** datasets
 - HN-WK, HN-YG, HN-FB, HN-FB-S
- Comparison with **12 baseline methods**
 - Methods for handling numeric literals
 - TransEA, MT-KGNN, KBLN, LiteralE
 - Methods for handling hyper-relational facts
 - NaLP, tNaLP, RAM, HINGE, NeuInfer, StarE, Hy-Transformer, GRAN

Link Prediction Results – Primary

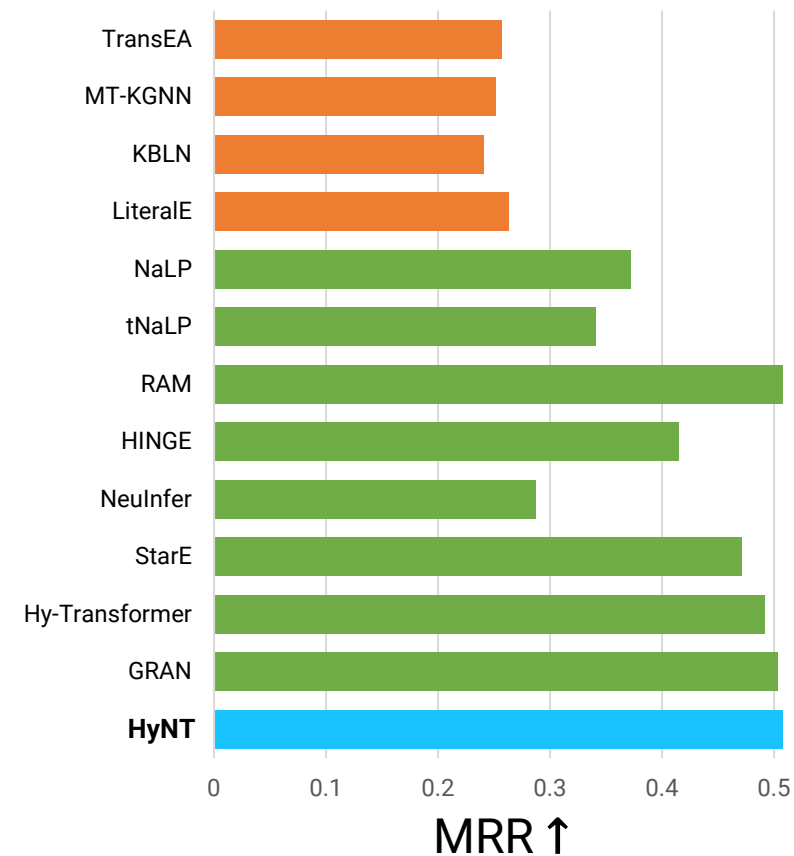
HN-WK



HN-YG

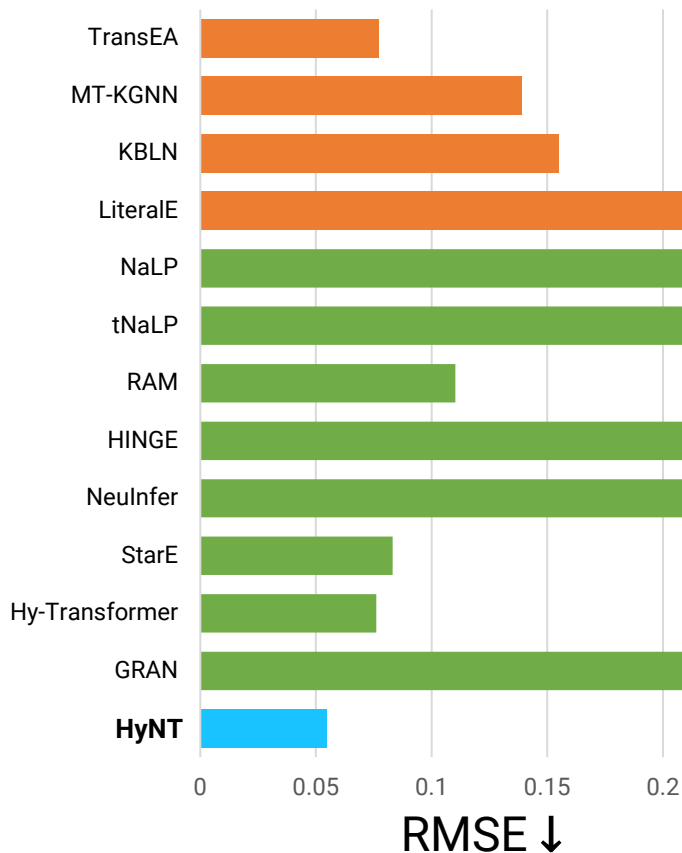


HN-FB-S

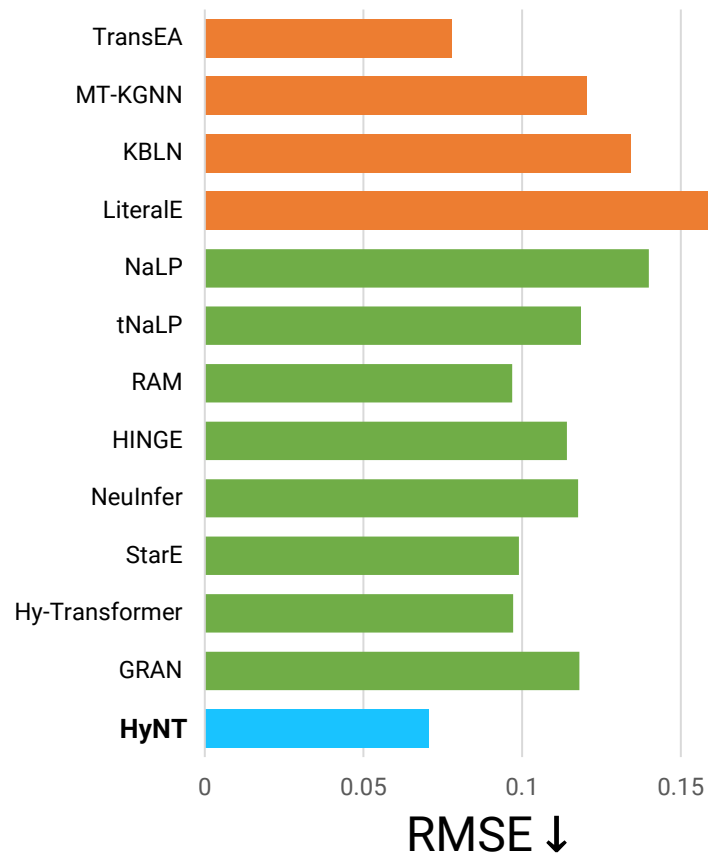


Numeric Value Prediction Results – Primary

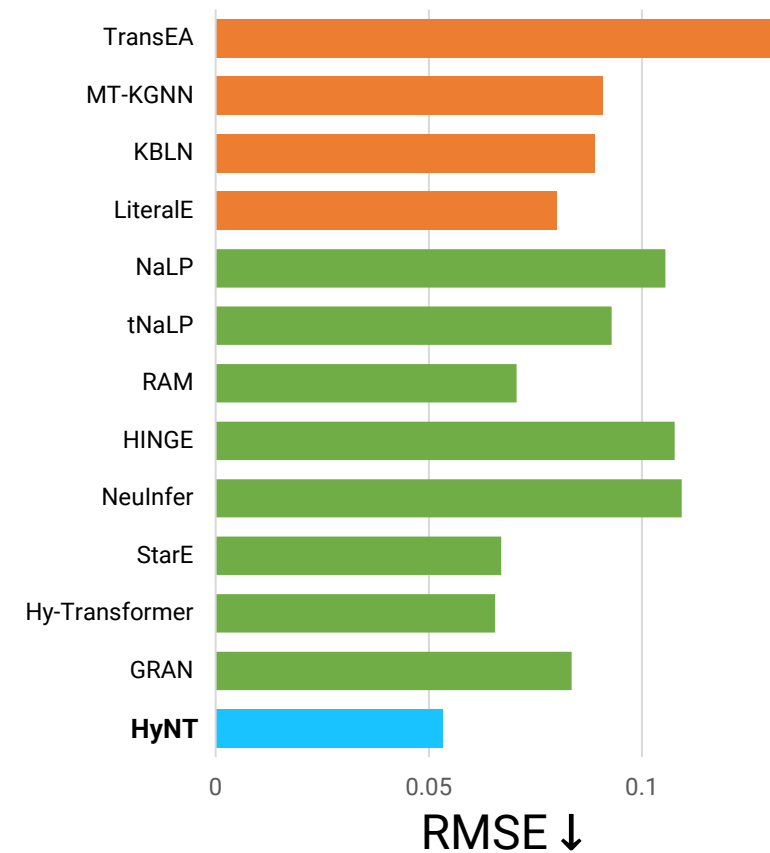
HN-WK



HN-YG



HN-FB-S



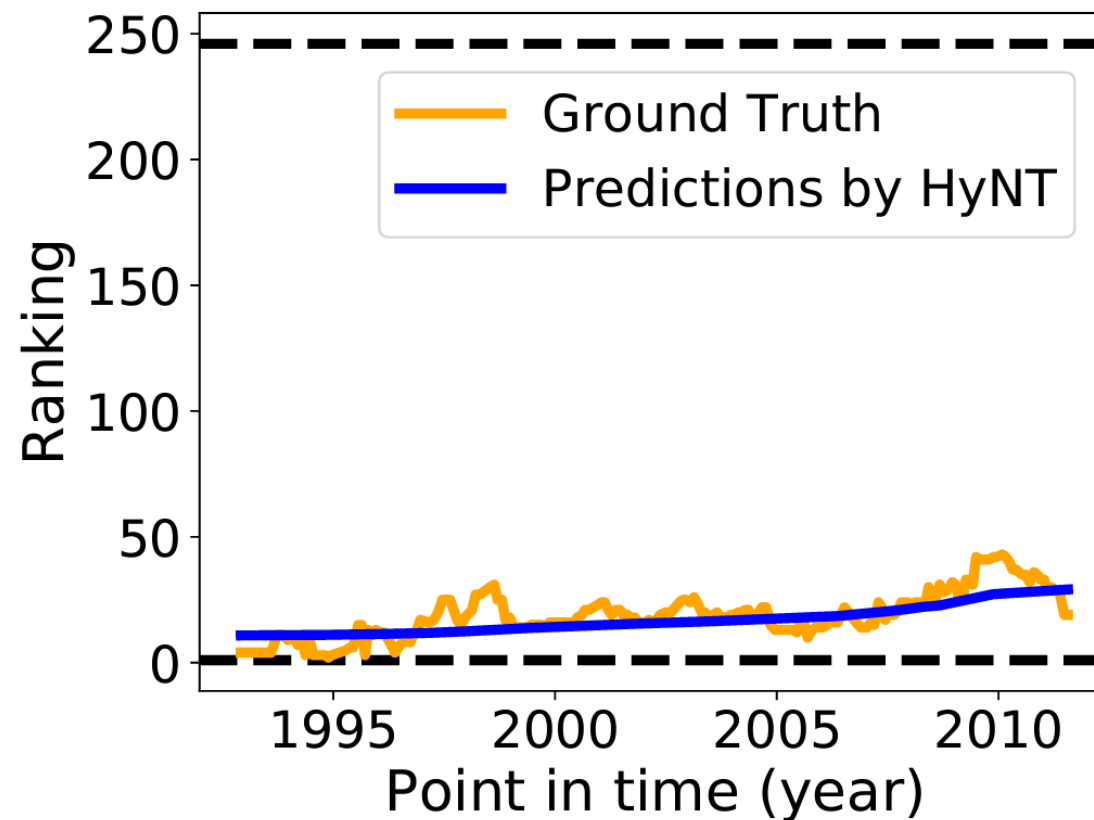
Target Values

((Sweden National team, ranking, ?), {(point in time, 1995)})
((Sweden National team, ranking, ?), {(point in time, 1996)})
((Sweden National team, ranking, ?), {(point in time, 1997)})
⋮

Visualization of the Predictions

(Sweden National Team, ranking, ?)

Target Values



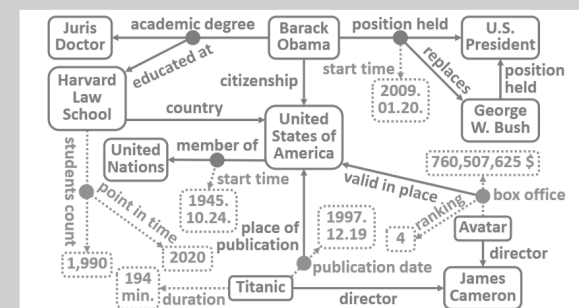
02 Conclusion & Future Work

- Hyper-relational and Numeric Knowledge Graphs (HN-KGs)
- Propose HyNT to solve link prediction, numeric value prediction, and relation prediction on HN-KGs
- HyNT significantly outperforms 12 different state-of-the-art methods
- Extend HyNT to inductive learning scenarios
 - New entities and relations appear at test time

Representation Learning on Hyper-Relational and Numeric Knowledge Graphs with Transformers

Chanyoung Chung[‡], Jaejun Lee[‡], and Joyce Jiyoung Whang^{*}

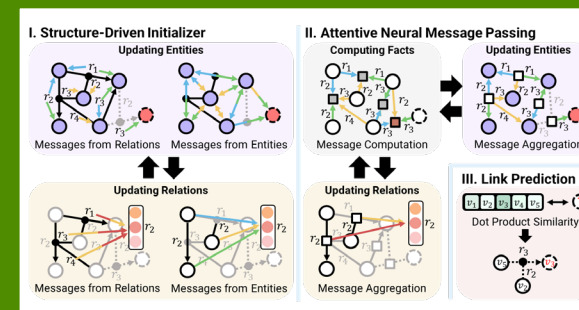
KDD 2023



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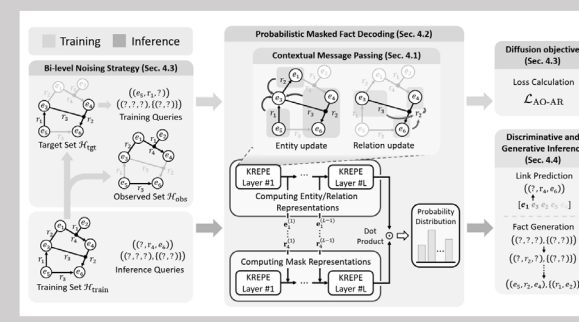
ICML 2025



Generative Representation Learning on Hyper-relational Knowledge Graphs via Masked Discrete Diffusion

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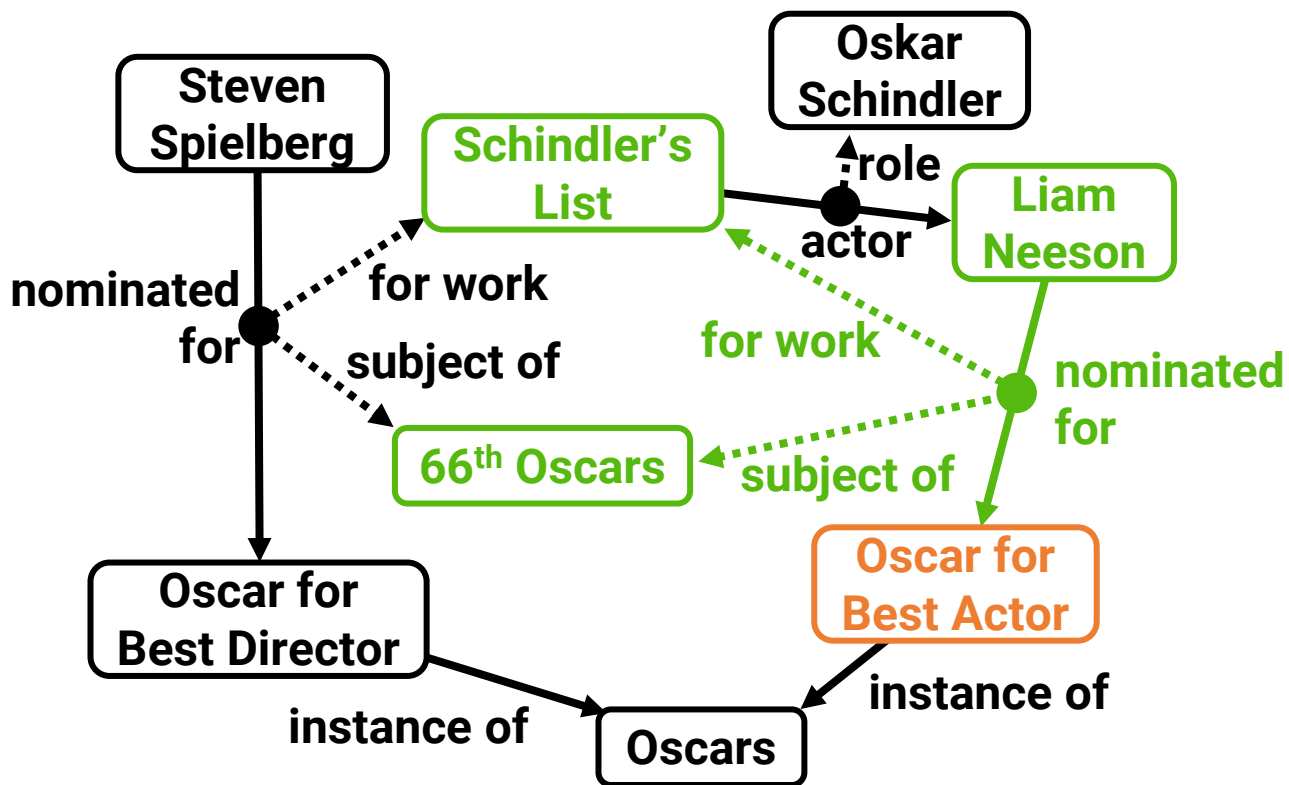
ICML 2026



03

Transductive Link Prediction on HKGs

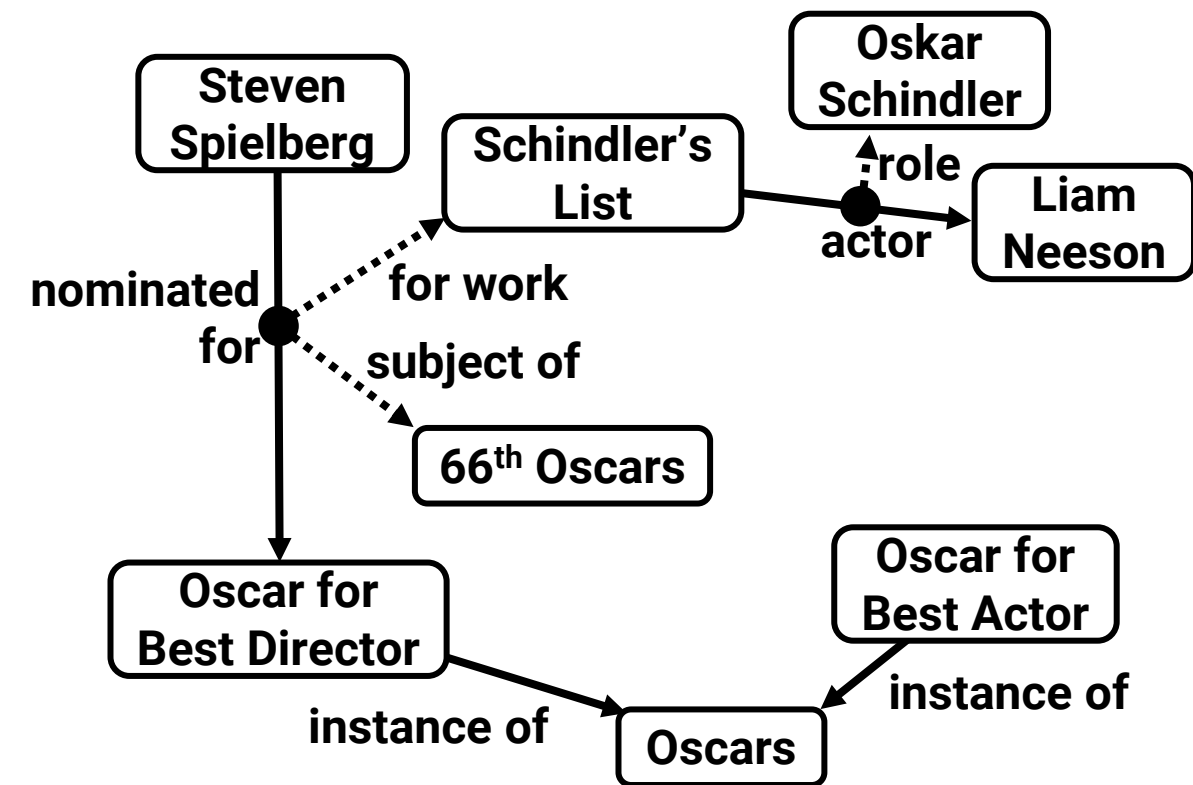
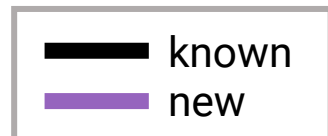
((Liam Neeson, nominated for, ?),
 {(for work, Schindler's List), (subject of, 66th Oscars)}))



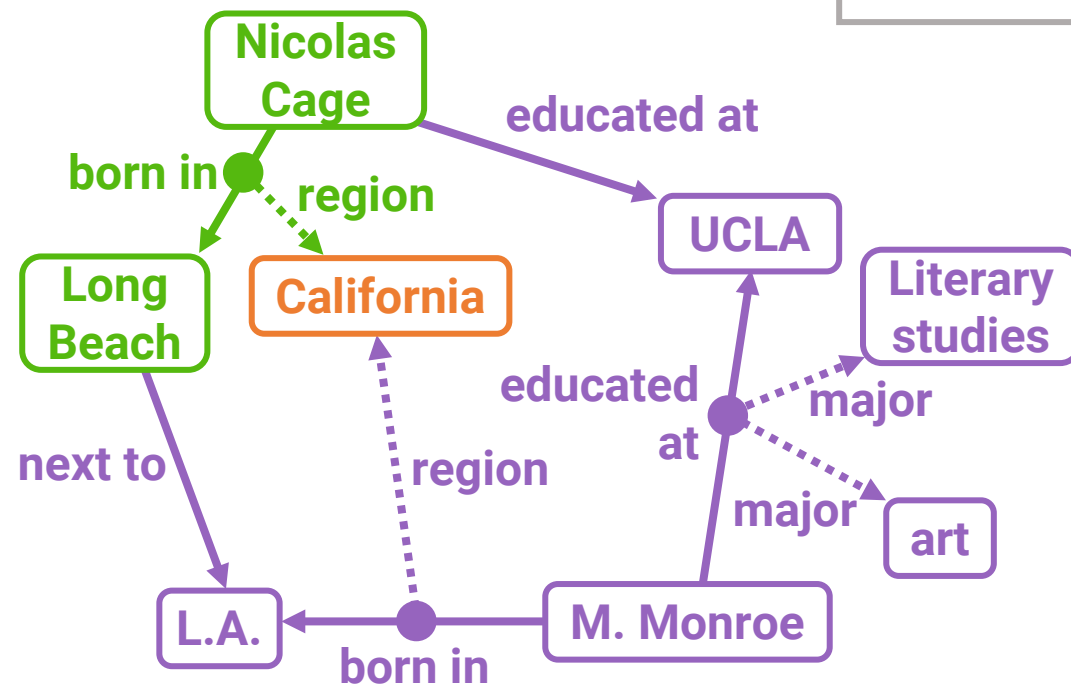
03

Inductive Link Prediction on HKGs

((Nicolas Cage, born in, Long Beach),
 {(region, ?)})



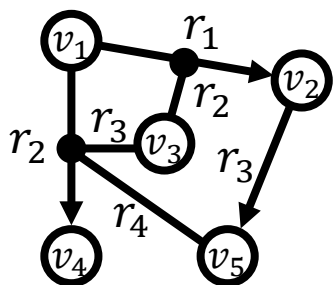
Training HKG



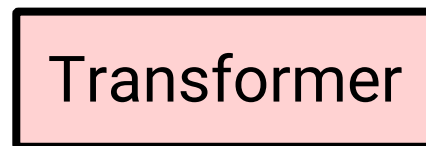
Inference HKG

03 Existing HKG Methods

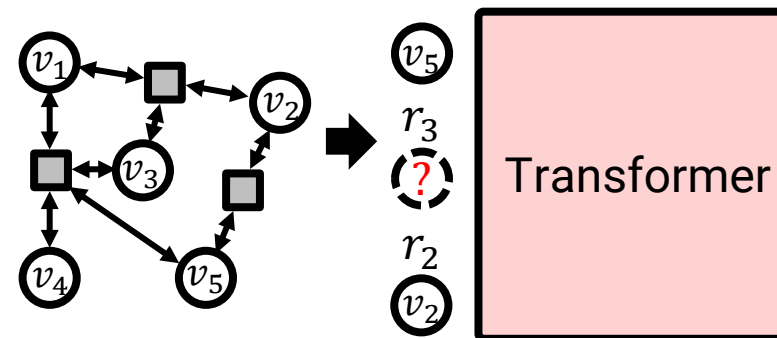
- Incorporate only limited structural information and fail to utilize HKG structures
 - Transformer-based methods: process each fact individually
 - GNN-based encoders: redundant / does not consider relations and positions of entities



Hyper-relational Knowledge Graph



Transformer-based Methods

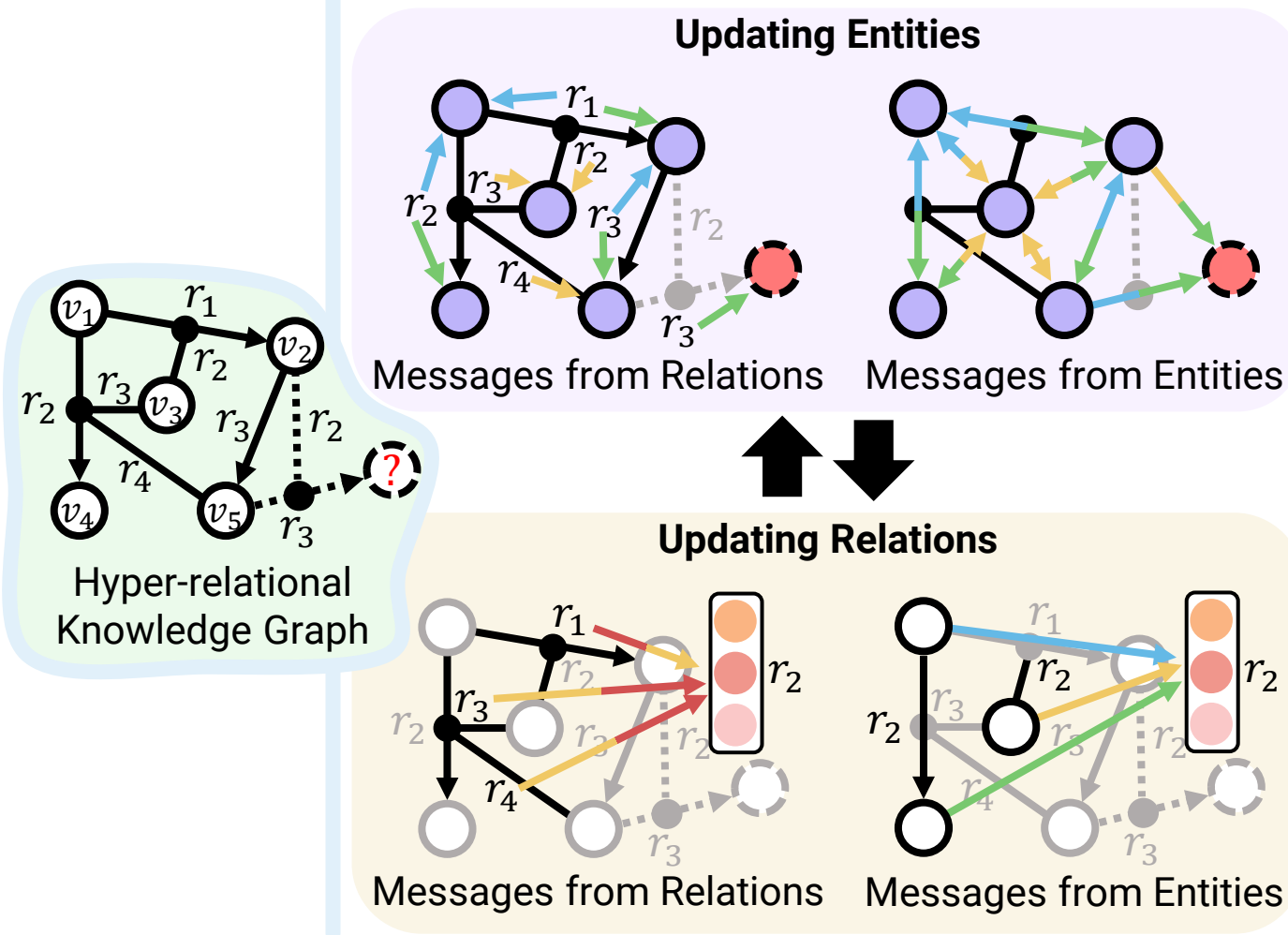


Existing GNN-based Methods

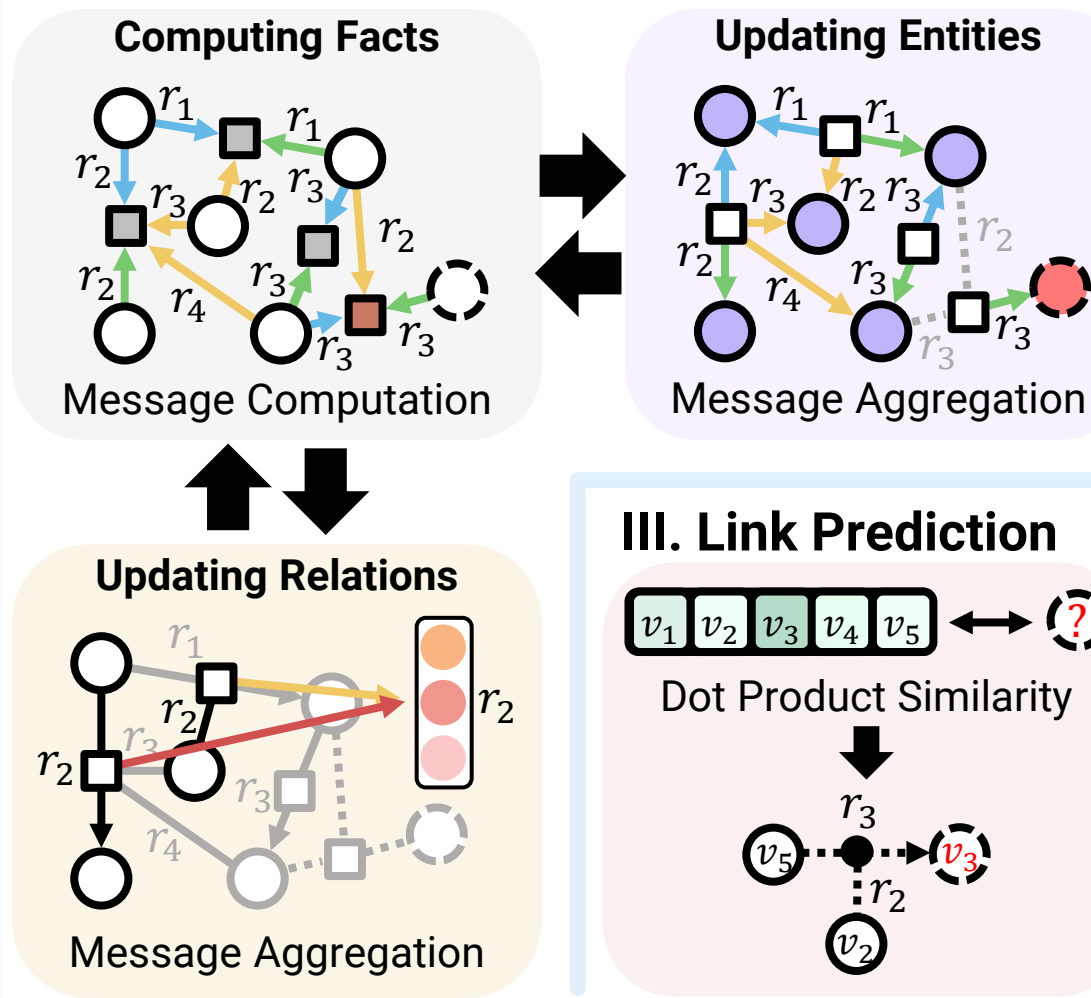
Inductive Inference	X	X
Structure Utilization	X	$\triangle$

MAYPL: Message Passing Framework for Hyper-Relational Knowledge Graph Representation Learning

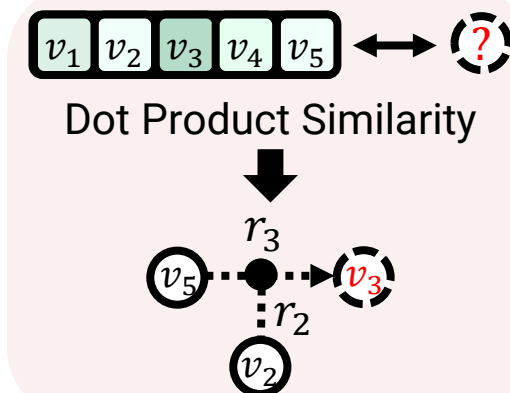
I. Structure-Driven Initializer



II. Attentive Neural Message Passing



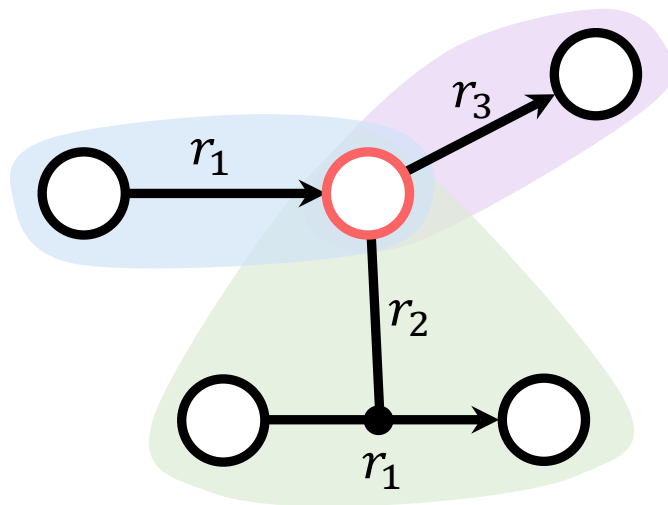
III. Link Prediction



- Exploits the **interconnections**, **co-occurrence**, and **positions** of entities and relations

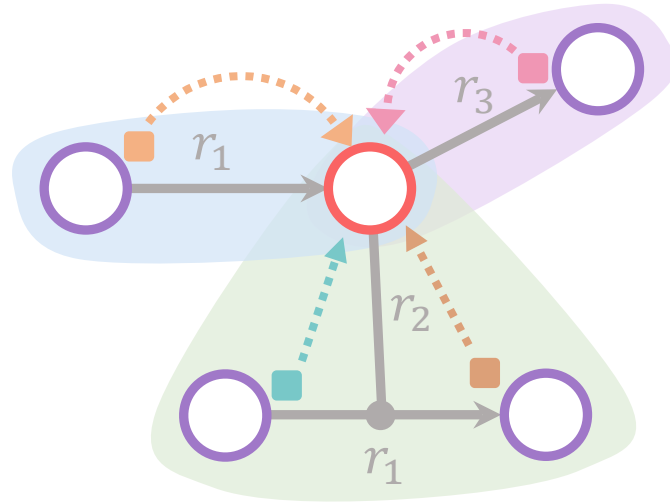
- Entity: Aggregate the messages of the co-occurring entities and incident relations

$$\tilde{\mathbf{v}}^{(\tilde{l})} = \frac{1}{\sum_{u \in \mathcal{V}_v} |\mathcal{H}_u \cap \mathcal{H}_v|} \sum_{u \in \mathcal{V}_v} \sum_{h \in \mathcal{H}_u \cap \mathcal{H}_v} \tilde{\mathbf{U}}_{\lambda_h(v)}^{(\tilde{l})} \tilde{\mathbf{W}}_{\lambda_h(u)}^{(\tilde{l})} \tilde{\mathbf{u}}^{(l-1)} + \frac{1}{|\mathcal{R}_v|} \sum_{r \in \mathcal{R}_v} \tilde{\mathbf{A}}_{\tau_r(v)}^{(\tilde{l})} \tilde{\mathbf{r}}^{(\tilde{l}-1)}$$



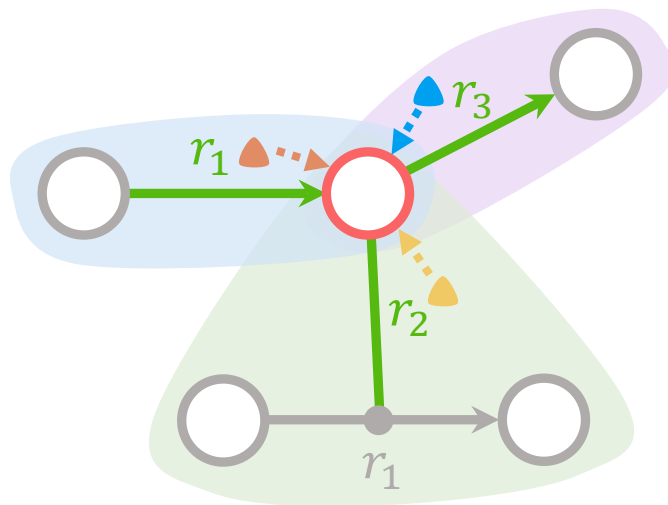
- Exploits the **interconnections**, **co-occurrence**, and **positions** of entities and relations
- Entity: Aggregate the messages of the **co-occurring entities** and incident relations

$$\tilde{\mathbf{v}}^{(\tilde{l})} = \frac{1}{\sum_{u \in \mathcal{V}_v} |\mathcal{H}_u \cap \mathcal{H}_v|} \sum_{u \in \mathcal{V}_v} \sum_{h \in \mathcal{H}_u \cap \mathcal{H}_v} \tilde{\mathbf{U}}_{\lambda_h(v)}^{(\tilde{l})} \tilde{\mathbf{W}}_{\lambda_h(u)}^{(\tilde{l})} \tilde{\mathbf{u}}^{(l-1)} + \frac{1}{|\mathcal{R}_v|} \sum_{r \in \mathcal{R}_v} \tilde{\mathbf{A}}_{\tau_r(v)}^{(\tilde{l})} \tilde{\mathbf{r}}^{(\tilde{l}-1)}$$



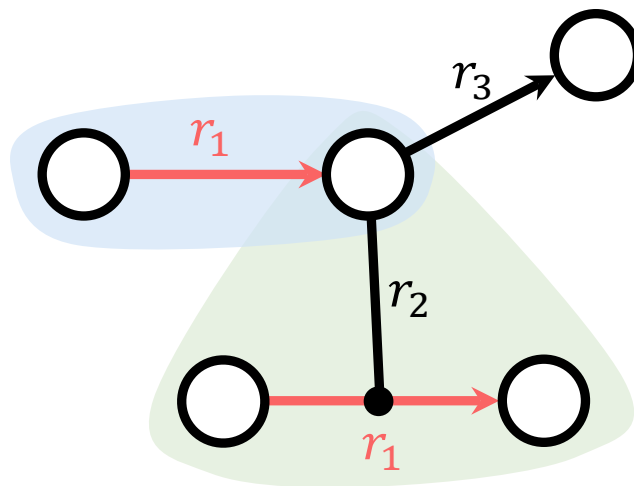
- Exploits the **interconnections**, **co-occurrence**, and **positions** of entities and relations
- Entity: Aggregate the messages of the co-occurring entities and **incident relations**

$$\tilde{\mathbf{v}}^{(\tilde{l})} = \frac{1}{\sum_{u \in \mathcal{V}_v} |\mathcal{H}_u \cap \mathcal{H}_v|} \sum_{u \in \mathcal{V}_v} \sum_{h \in \mathcal{H}_u \cap \mathcal{H}_v} \tilde{\mathbf{U}}_{\lambda_h(v)}^{(\tilde{l})} \tilde{\mathbf{W}}_{\lambda_h(u)}^{(\tilde{l})} \tilde{\mathbf{u}}^{(l-1)} + \frac{1}{|\mathcal{R}_v|} \sum_{r \in \mathcal{R}_v} \tilde{\mathbf{A}}_{\tau_r(v)}^{(\tilde{l})} \tilde{\mathbf{r}}^{(\tilde{l}-1)}$$



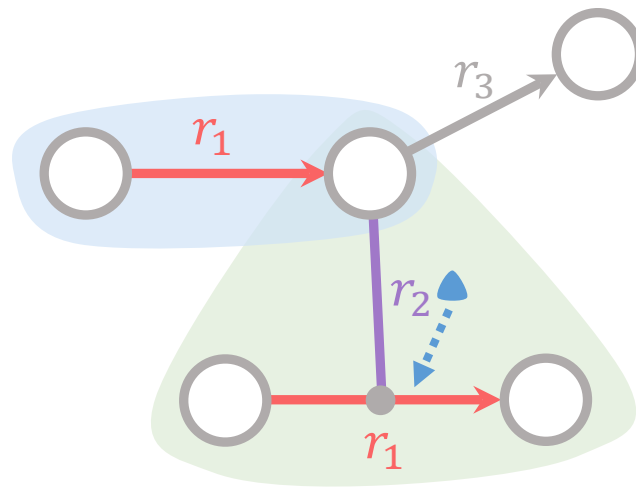
- Exploits the **interconnections**, **co-occurrence**, and **positions** of entities and relations
- Relation: Aggregate the messages of the co-occurring relations and incident entities

$$\tilde{\mathbf{r}}^{(\tilde{l})} = \frac{1}{\sum_{y \in \mathcal{R}_r} |\mathcal{H}_y \cap \mathcal{H}_r|} \sum_{y \in \mathcal{R}_r} \sum_{h \in \mathcal{H}_y \cap \mathcal{H}_r} \hat{\mathbf{U}}_{\lambda_h(r)}^{(\tilde{l})} \hat{\mathbf{W}}_{\lambda_h(y)}^{(\tilde{l})} \tilde{\mathbf{y}}^{(l-1)} + \frac{1}{|\mathcal{V}_r|} \sum_{v \in \mathcal{V}_r} \hat{\mathbf{A}}_{\tau_r(v)}^{(\tilde{l})} \tilde{\mathbf{v}}^{(l-1)}$$



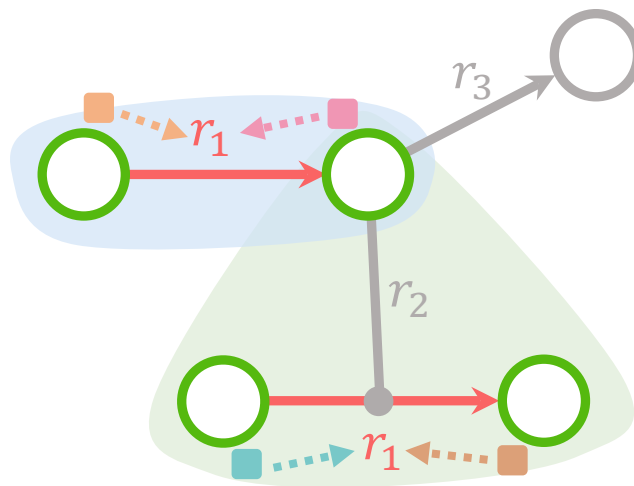
- Exploits the **interconnections**, **co-occurrence**, and **positions** of entities and relations
- Relation: Aggregate the messages of the **co-occurring relations** and incident entities

$$\tilde{\mathbf{r}}^{(\tilde{l})} = \frac{1}{\sum_{y \in \mathcal{R}_r} |\mathcal{H}_y \cap \mathcal{H}_r|} \sum_{y \in \mathcal{R}_r} \sum_{h \in \mathcal{H}_y \cap \mathcal{H}_r} \hat{\mathbf{U}}_{\lambda_h(r)}^{(\tilde{l})} \hat{\mathbf{W}}_{\lambda_h(y)}^{(\tilde{l})} \tilde{\mathbf{y}}^{(l-1)} + \frac{1}{|\mathcal{V}_r|} \sum_{v \in \mathcal{V}_r} \hat{\mathbf{A}}_{\tau_r(v)}^{(\tilde{l})} \tilde{\mathbf{v}}^{(l-1)}$$



- Exploits the **interconnections**, **co-occurrence**, and **positions** of entities and relations
- Relation: Aggregate the messages of the co-occurring relations and **incident entities**

$$\tilde{\mathbf{r}}^{(\tilde{l})} = \frac{1}{\sum_{y \in \mathcal{R}_r} |\mathcal{H}_y \cap \mathcal{H}_r|} \sum_{y \in \mathcal{R}_r} \sum_{h \in \mathcal{H}_y \cap \mathcal{H}_r} \hat{\mathbf{U}}_{\lambda_h(r)}^{(\tilde{l})} \hat{\mathbf{W}}_{\lambda_h(y)}^{(\tilde{l})} \tilde{\mathbf{y}}^{(l-1)} + \frac{1}{|\mathcal{V}_r|} \sum_{v \in \mathcal{V}_r} \hat{\mathbf{A}}_{\tau_r(v)}^{(\tilde{l})} \tilde{\mathbf{v}}^{(l-1)}$$

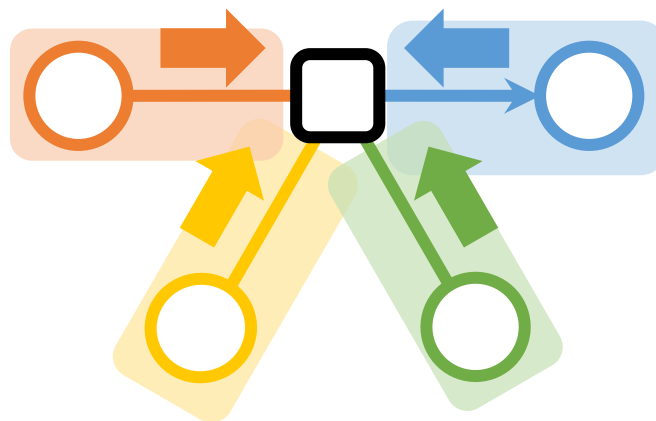


Attentive Neural Message Passing

- Updates entity and relation representations by attentively aggregating facts' messages
- Computes a fact's message by considering which entities and relations comprise the fact
 - Decompose each fact as a set of its relation-entity pairs
 - compute a pair's message by considering the **entities** and **relations** and their **positions** within the fact
- Compute a fact's message by aggregating messages of its **relation-entity pairs**

$$\mathbf{p}^{(l)} = P_{\lambda_h(v)}^{(l)} \left(\left(W_{\lambda_h(v)}^{(l)} \mathbf{v}^{(l-1)} \right) \odot \left(U_{\lambda_h(v)}^{(l)} \mathbf{r}^{(l-1)} \right) \right)$$

$$\mathbf{h}^{(l)} = \frac{1}{|(\mathcal{R} \times \mathcal{V})_h|} \sum_{p \in (\mathcal{R} \times \mathcal{V})_h} \mathbf{p}^{(l)}$$



Attentive Neural Message Passing

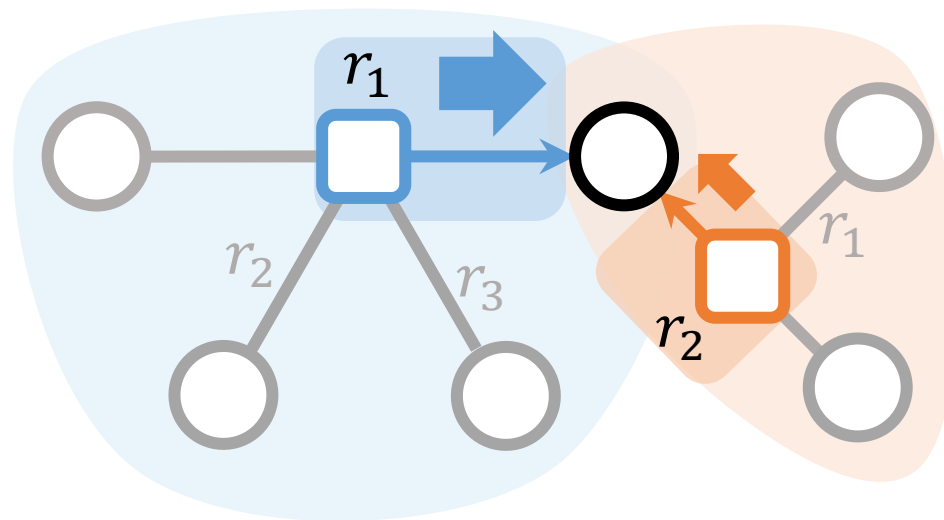
- Updates an entity representation by considering which fact the entity belongs to

- Considers pairs of **incident relations** and the **corresponding facts**

- $$\mathbf{q}^{(l)} = \mathbf{Q}_{\lambda_h(v)}^{(l)} \left(\mathbf{h}^{(l)} \odot \left(\mathbf{A}_{\lambda_h(v)}^{(l)} \mathbf{r}^{(l-1)} \right) \right)$$

- Attentively aggregate relation-fact pairs

- $$\mathbf{v}^{(l)} = \sum_{q \in (\mathcal{R} \times \mathcal{H})_v} \alpha_{q,v}^{(l)} \mathbf{B}^{(l)} \mathbf{q}^{(l)}, \text{ where } \alpha_{q,v}^{(l)} = \frac{\exp(\mathbf{a}^{(l)} \cdot \sigma(\mathbf{Q}^{(l)} \mathbf{v}^{(l-1)} + \mathbf{K}^{(l)} \mathbf{q}^{(l)}))}{\sum_{k \in (\mathcal{R} \times \mathcal{H})_v} \exp(\mathbf{a}^{(l)} \cdot \sigma(\mathbf{Q}^{(l)} \mathbf{v}^{(l-1)} + \mathbf{K}^{(l)} \mathbf{k}^{(l)}))}$$



03

Attentive Neural Message Passing

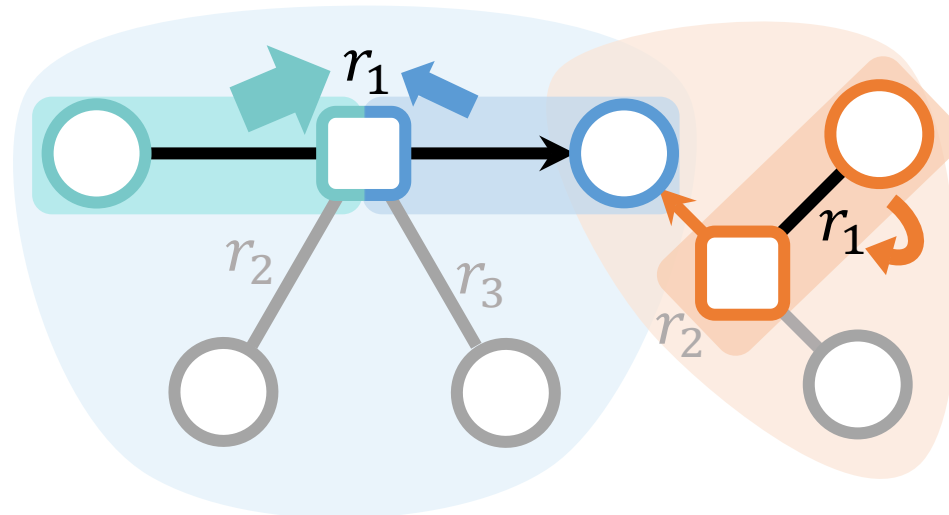
- Updates a relation representation by considering which fact the relation belongs to

- Considers pairs of **incident entities** and the **corresponding facts**

- $\bar{\mathbf{q}}^{(l)} = \bar{\mathbf{Q}}_{\lambda_h(r)}^{(l)} \left(\mathbf{h}^{(l)} \odot \left(\bar{\mathbf{A}}_{\lambda_h(v)}^{(l)} \mathbf{v}^{(l-1)} \right) \right)$

- Attentively aggregate relation-fact pairs

- $\mathbf{r}^{(l)} = \sum_{\bar{q} \in (\mathcal{V} \times \mathcal{H})_r} \bar{\alpha}_{\bar{q},r}^{(l)} \bar{\mathbf{B}}^{(l)} \bar{\mathbf{q}}^{(l)}$, where $\bar{\alpha}_{\bar{q},r}^{(l)} = \frac{\exp(\bar{\mathbf{a}}^{(l)} \cdot \sigma(\bar{\mathbf{Q}}^{(l)} \mathbf{r}^{(l-1)} + \bar{\mathbf{K}}^{(l)} \bar{\mathbf{q}}^{(l)}))}{\sum_{\bar{k} \in (\mathcal{V} \times \mathcal{H})_r} \exp(\bar{\mathbf{a}}^{(l)} \cdot \sigma(\bar{\mathbf{Q}}^{(l)} \mathbf{r}^{(l-1)} + \bar{\mathbf{K}}^{(l)} \bar{\mathbf{k}}^{(l)}))}$



03 Link Prediction on HKGs

- Structure-driven initializer

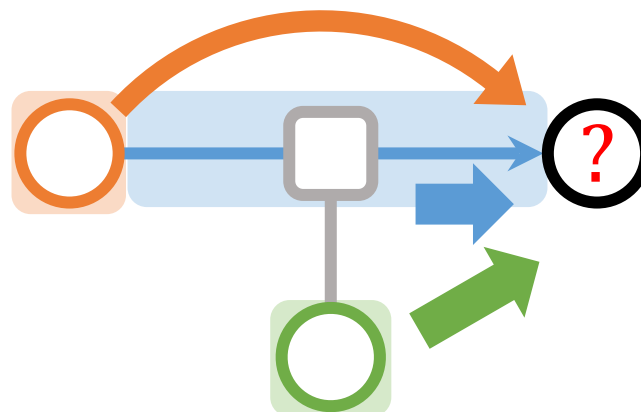
- $\tilde{\mathbf{x}}^{(\tilde{l})} = \sum_{v \in \mathcal{V}_x} \tilde{\mathbf{U}}_{\lambda_h(x)}^{(\tilde{l})} \tilde{\mathbf{W}}_{\lambda_h(v)}^{(\tilde{l})} \tilde{\mathbf{v}}^{(l-1)} + \tilde{\mathbf{A}}_{\tau_r(v)}^{(\tilde{l})} \tilde{\mathbf{r}}^{(l-1)}$

- Attentive Neural Message Passing

- $\mathbf{x}^{(l)} = \mathbf{B}^{(l)} \left(\mathbf{Q}_{\lambda_h(x)}^{(l)} \left(\mathbf{h}^{(l)} \odot \left(\mathbf{A}_{\lambda_h(x)}^{(l)} \mathbf{r}^{(l-1)} \right) \right) \right)$

- Compute the dot product similarity between $\mathbf{x}^{(L)}$ and each entity representation

- MAYPL predicts the missing entity as the entity with the highest similarity



03 Link Prediction on HKGs

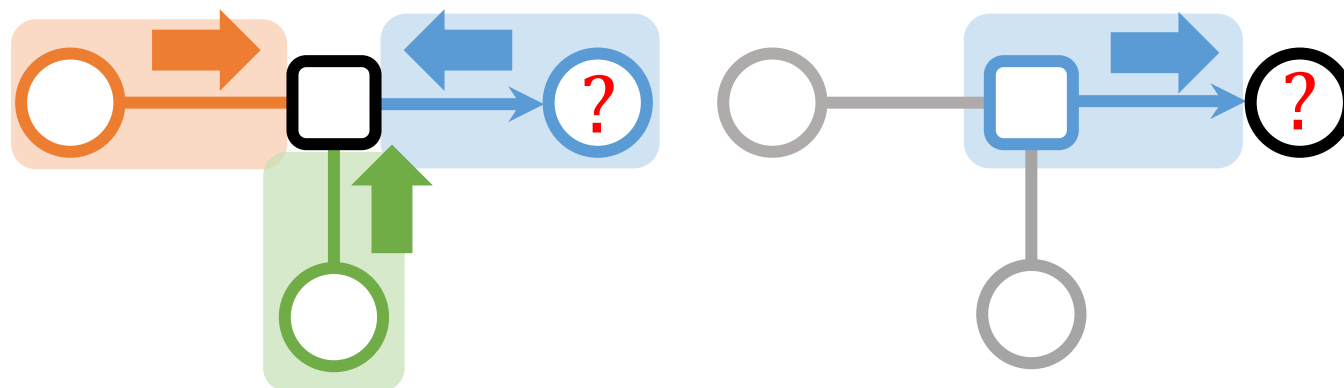
- Structure-driven initializer

$$\tilde{\mathbf{x}}^{(\tilde{l})} = \sum_{v \in \mathcal{V}_x} \tilde{\mathbf{U}}_{\lambda_h(x)}^{(\tilde{l})} \tilde{\mathbf{W}}_{\lambda_h(v)}^{(\tilde{l})} \tilde{\mathbf{v}}^{(l-1)} + \tilde{\mathbf{A}}_{\tau_r(v)}^{(\tilde{l})} \tilde{\mathbf{r}}^{(\tilde{l}-1)}$$

- Attentive Neural Message Passing

$$\mathbf{x}^{(l)} = \mathbf{B}^{(l)} \left(\mathbf{Q}_{\lambda_h(x)}^{(l)} \left(\mathbf{h}^{(l)} \odot \left(\mathbf{A}_{\lambda_h(x)}^{(l)} \mathbf{r}^{(l-1)} \right) \right) \right)$$

- Compute the dot product similarity between $\mathbf{x}^{(L)}$ and each entity representation
 - MAYPL predicts the missing entity as the entity with the highest similarity



03

Link Prediction on HKGs

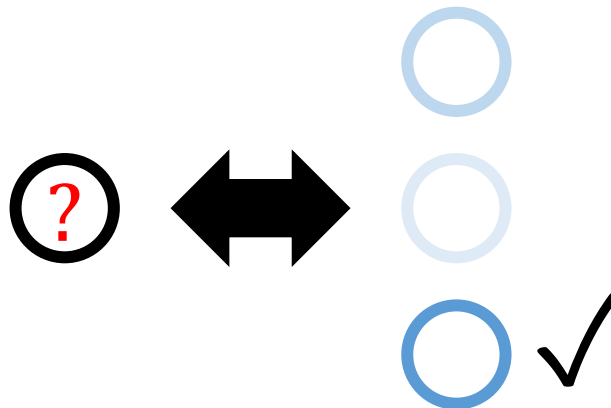
- Structure-driven initializer

- $\tilde{\mathbf{x}}^{(\tilde{l})} = \sum_{v \in \mathcal{V}_x} \tilde{\mathbf{U}}_{\lambda_h(x)}^{(\tilde{l})} \tilde{\mathbf{W}}_{\lambda_h(v)}^{(\tilde{l})} \tilde{\mathbf{v}}^{(l-1)} + \tilde{\mathbf{A}}_{\tau_r(v)}^{(\tilde{l})} \tilde{\mathbf{r}}^{(\tilde{l}-1)}$

- Attentive Neural Message Passing

- $\mathbf{x}^{(l)} = \mathbf{B}^{(l)} \left(\mathbf{Q}_{\lambda_h(x)}^{(l)} \left(\mathbf{h}^{(l)} \odot \left(\mathbf{A}_{\lambda_h(x)}^{(l)} \mathbf{r}^{(l-1)} \right) \right) \right)$

- Compute the dot product similarity between $\mathbf{x}^{(L)}$ and each entity representation
 - MAYPL predicts the missing entity as the entity with the highest similarity

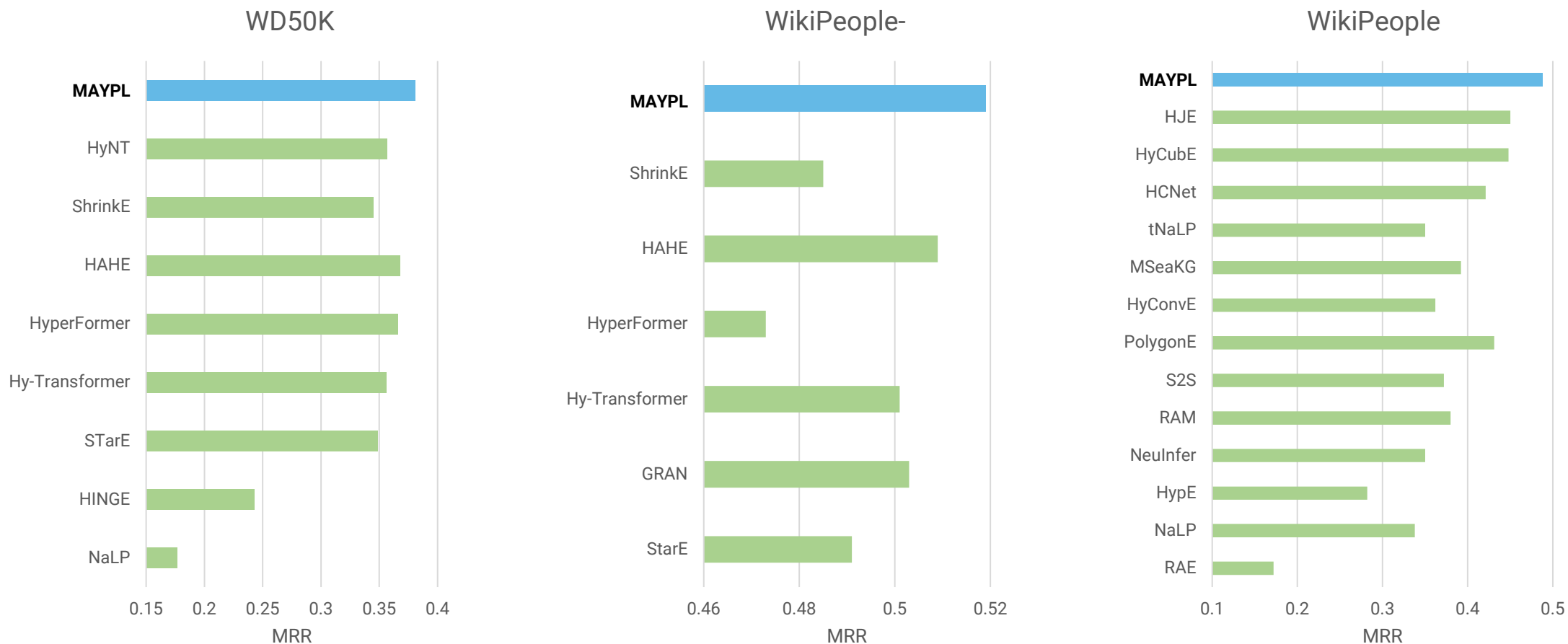


03 Experimental Results

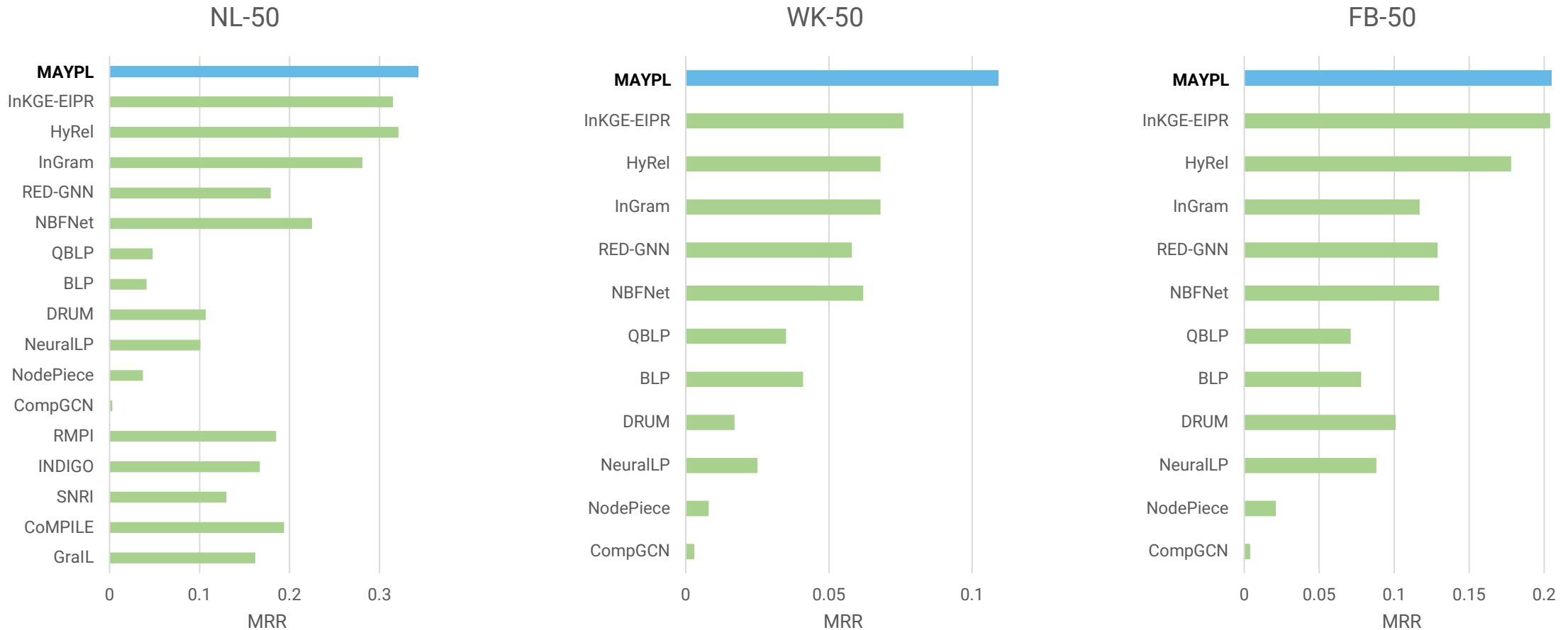
- Datasets
 - **3 Transductive HKG** datasets: WD50K, WikiPeople⁻, WikiPeople
 - **12 Inductive KG** datasets from InGram (e.g., NL-100, WK-100, FB-100)
 - **4 Inductive HKG** datasets: WD20K(100)v1, WD20K(100)v2, WP-IND, MFB-IND
- Baselines
 - **41 knowledge graph completion methods**, compared with different baseline methods on different datasets

03

Transductive Link Prediction on HKGs

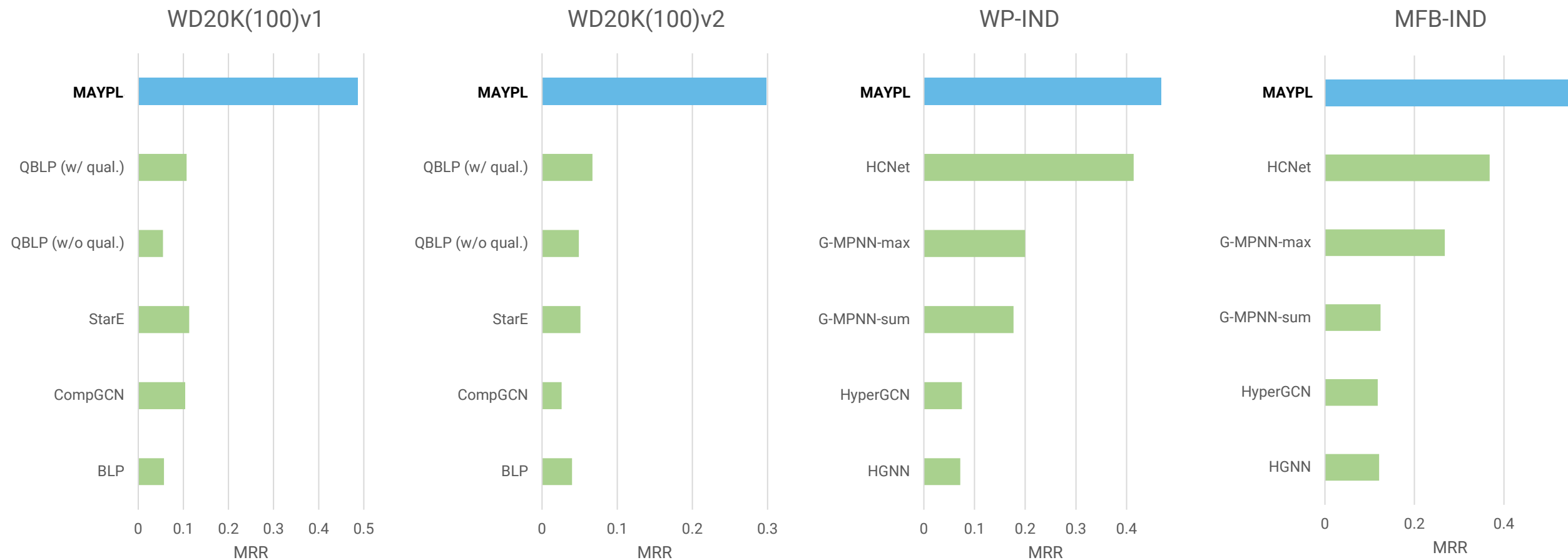


Inductive Link Prediction on KGs



03

Inductive Link Prediction on HKGs



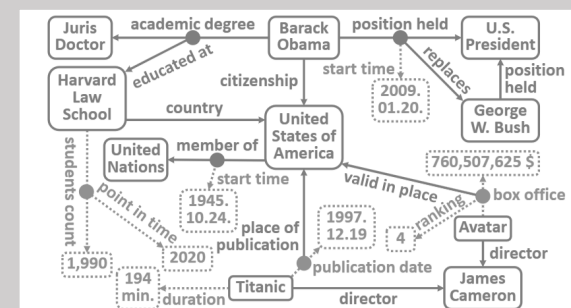
03 Conclusion

- Employing an **HKG's structure is crucial** for HKG reasoning
 - Thoroughly learning and exploiting the structure of an HKG is necessary and sufficient for learning representations on HKGs
- Propose **MAYPL**, the first structure-oriented representation learning method for HKGs
 - Can be applied in both transductive and inductive settings
- MAYPL can make inductive inferences with new entities and relations by **learning how to compute representations based solely on the structure** of a given HKG
- MAYPL outperforms **41 different methods on 19 benchmark datasets** in varied settings
 - Transductive link prediction on HKGs, inductive link prediction on KGs, and inductive link prediction on HKGs

Representation Learning on Hyper-Relational and Numeric Knowledge Graphs with Transformers

Chanyoung Chung[‡], Jaejun Lee[‡], and Joyce Jiyoung Whang^{*}

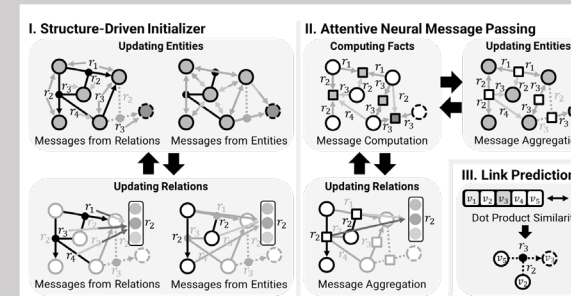
KDD 2023



Structure Is All You Need: Structural Representation Learning on Hyper-Relational Knowledge Graphs

Jaejun Lee and Joyce Jiyoung Whang^{*}

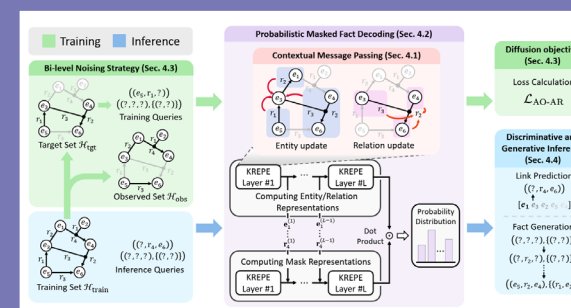
ICML 2025



Generative Representation Learning on Hyper-relational Knowledge Graphs via Masked Discrete Diffusion

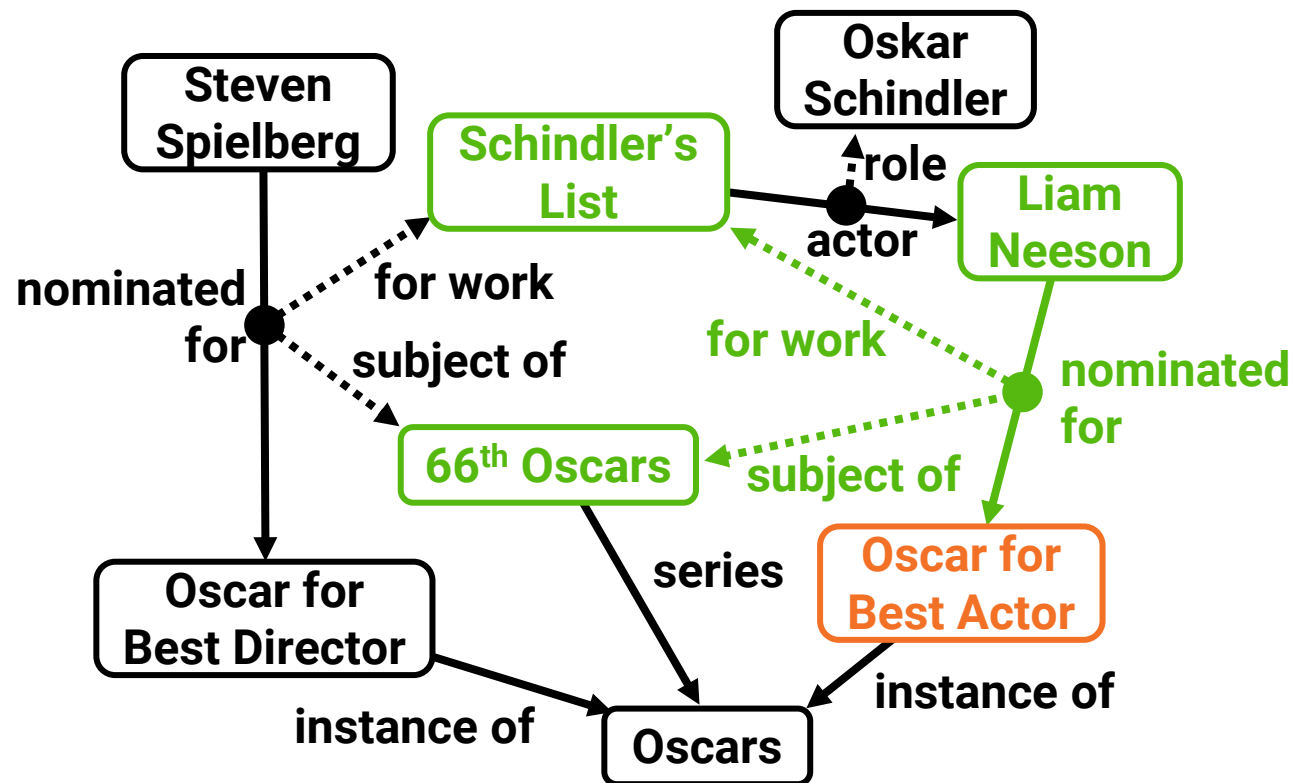
Jaejun Lee, Seheon Kim, and Joyce Jiyoung Whang^{*}

ICML 2026



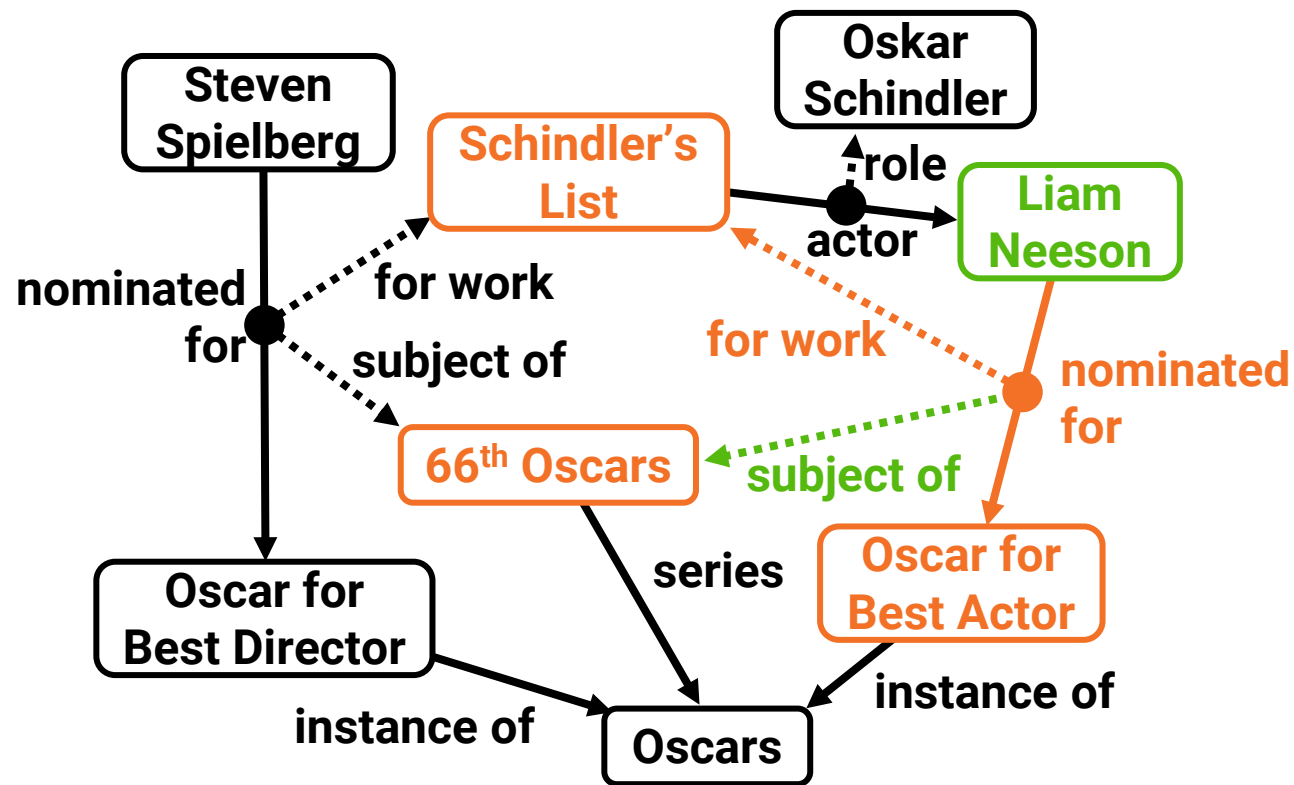
Link Prediction on HKGs

((Liam Neeson, nominated for, ?),
 {(for work, Schindler's List), (subject of, 66th Oscars)}))



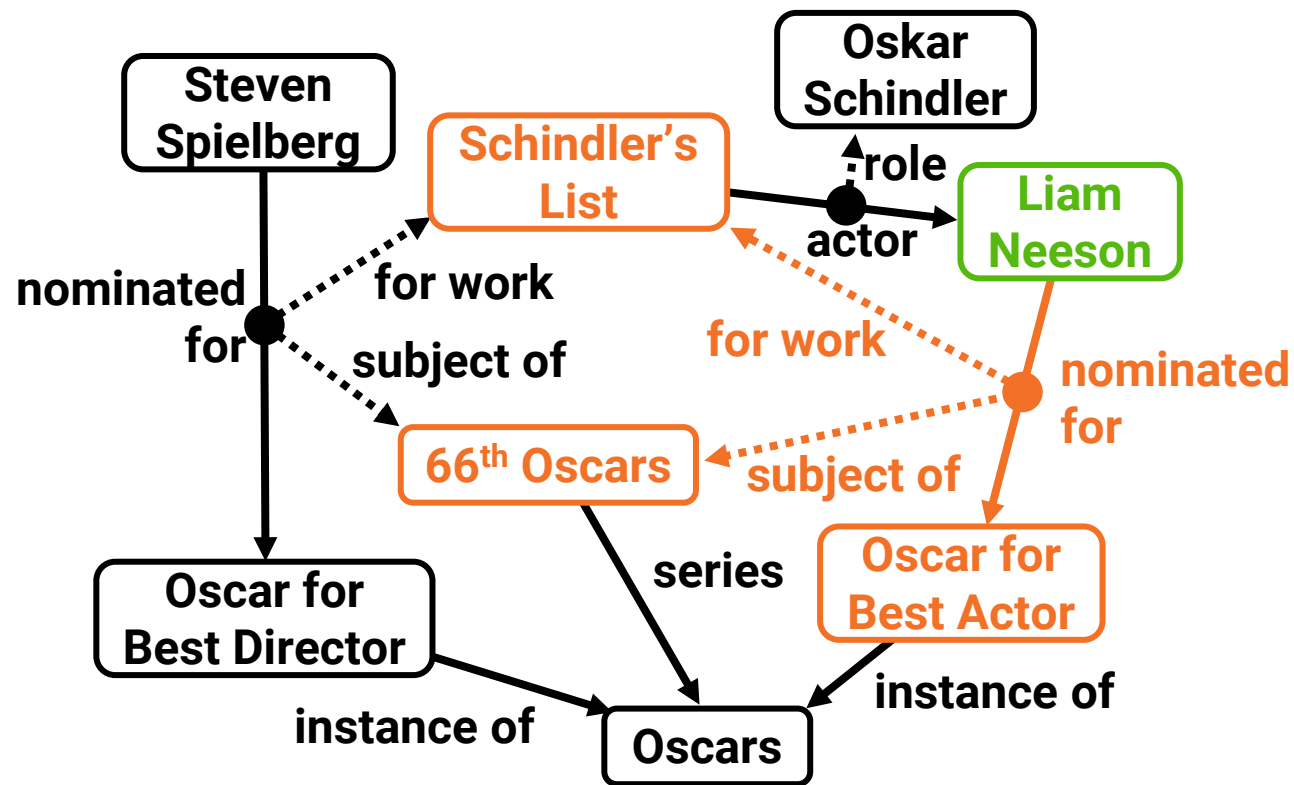
Fact Generation on HKGs

$((\text{Liam Neeson}, \text{?}, \text{?}), (\text{subject of}, \text{?}))$
 $\{(\text{?}, \text{?}), (\text{subject of}, \text{?})\}$



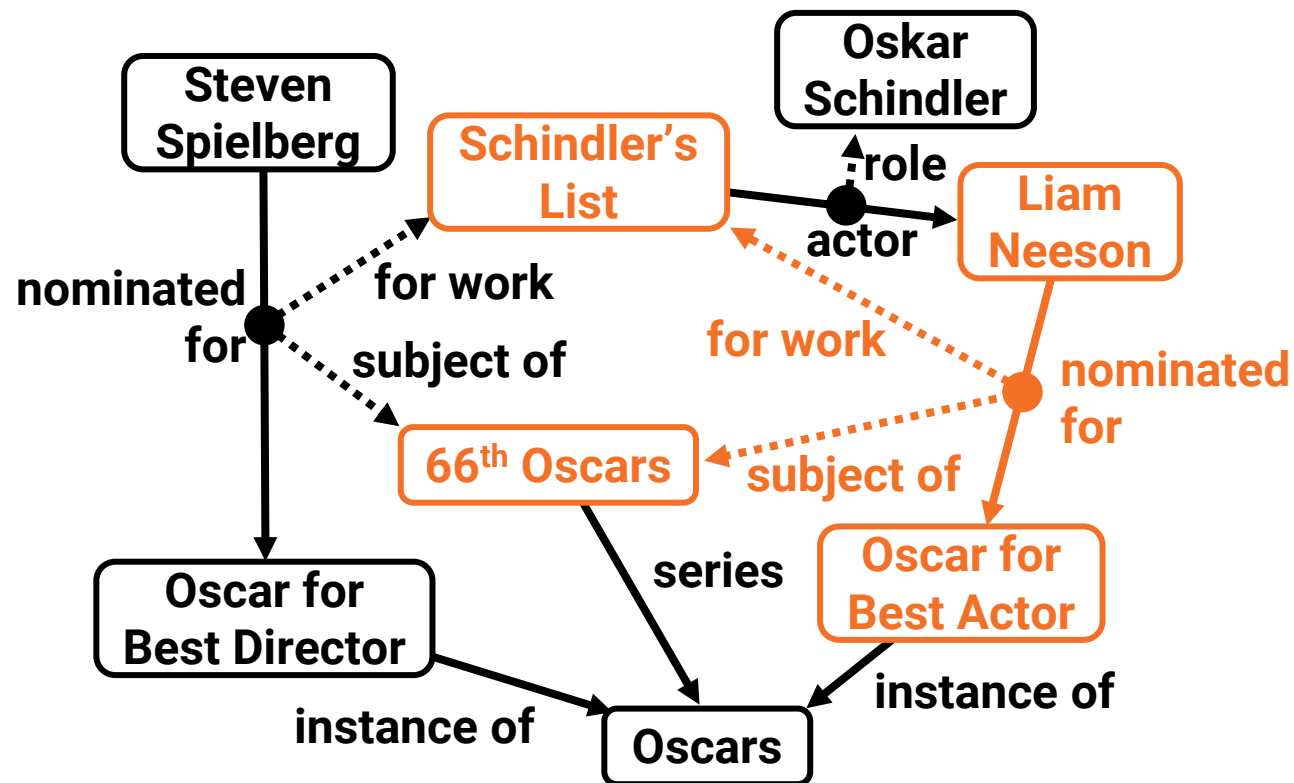
Fact Generation on HKGs

((Liam Neeson, ?, , ?),
{(?, , ?), (?, , ?)})

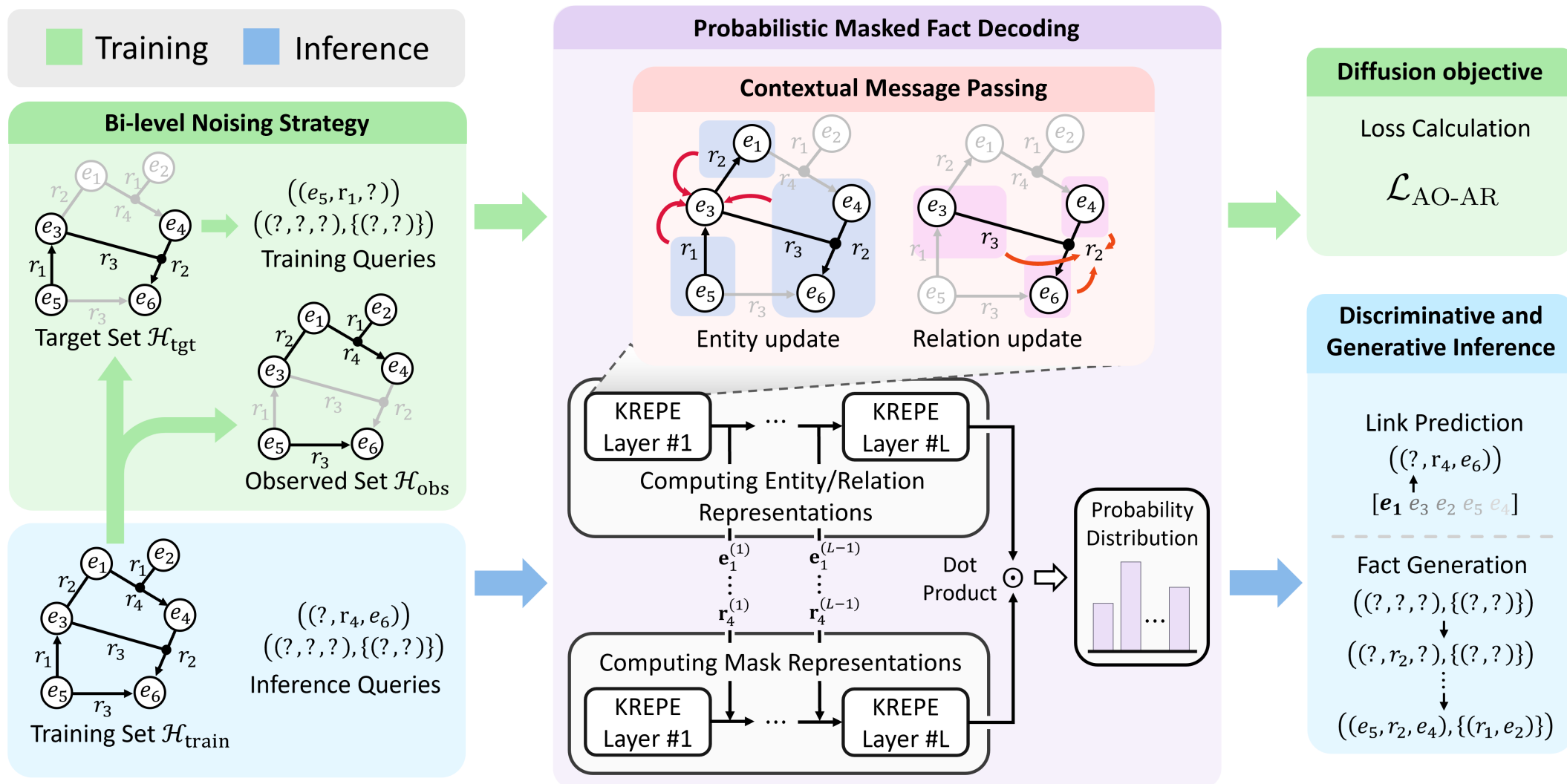


Fact Generation on HKGs

((? , ? , ?), (? , ?))
 {(? , ?), (? , ?)}

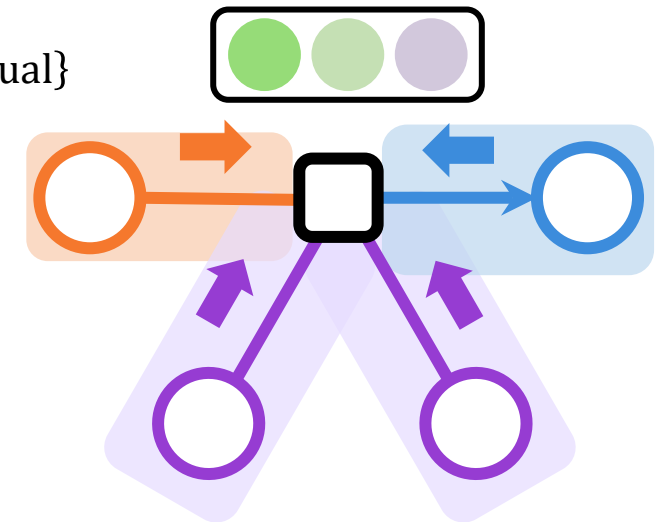


KREPE: Contextual HKG Representation Learning via Masked Discrete Diffusion



04 Contextual Message Passing

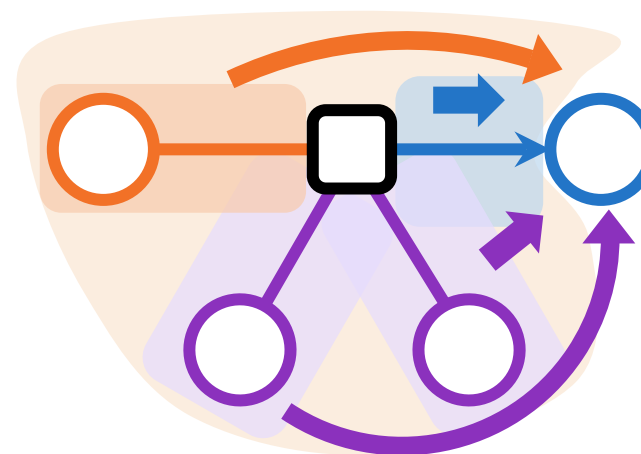
- Updates entity and relation representations by **aggregating messages from the facts they belong to** while explicitly **excluding their own information** from those facts' messages
 - Forces the representations to rely solely on the surrounding **context** in the facts
- The representation of a fact $\xi = ((h, r, t), Q_\xi)$ is obtained via the **sum of the representations of its constituent relation-entity pairs**
 - A fact is decomposed into a set of its constituent relation-entity pairs
 - The representation of a relation-entity pair (p, e) with role $\rho \in \{\text{head, tail, qual}\}$ is computed by: $\mathbf{z}_{(p,e)}^{(l)} = \mathbf{W}_\rho^{(l)} [\mathbf{p}^{(l-1)}; \mathbf{e}^{(l-1)}]$
 - $\mathbf{z}_\xi^{(l)} = \mathbf{z}_{(r,h)}^{(l)} + \mathbf{z}_{(r,t)}^{(l)} + \sum_{(k,v) \in Q_\xi} \mathbf{z}_{(k,v)}^{(l)}$



04 Contextual Message Passing

- Updates entity and relation representations by **aggregating messages from the facts they belong to** while explicitly **excluding their own information** from those facts' messages
 - Forces the representations to rely solely on the surrounding **context** in the facts
- To update a component within a fact, we compute a **context message** that encodes **all the other components in the fact**
 - The context messages from fact ξ to its components e and p in the constituent air (p, e) are computed by:

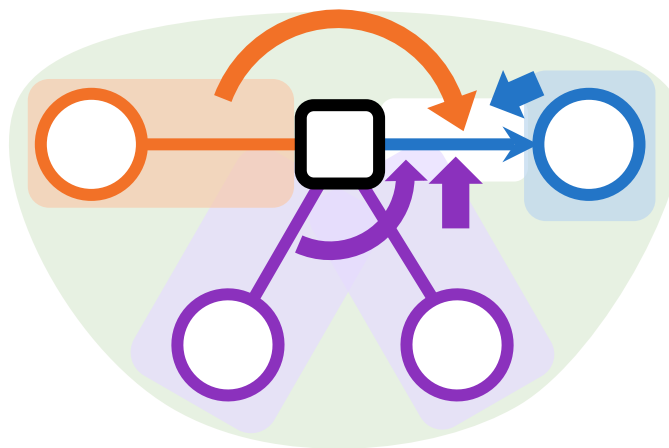
- $\mathbf{m}_{\xi \rightarrow (p,e)}^{(l)} = \text{MLP}^{(l)} \left(\frac{\mathbf{z}_{\xi}^{(l)} - \mathbf{z}_{(p,e)}^{(l)}}{n_{\xi} + 1} \right)$
- $\mathbf{m}_{\xi \rightarrow e}^{(l)} = \mathbf{W}_{\text{ENT}}^{(l)} \left[\mathbf{m}_{\xi \rightarrow (p,e)}^{(l)}; \mathbf{p}^{(l-1)} \right]$
- $\mathbf{m}_{\xi \rightarrow p}^{(l)} = \mathbf{W}_{\text{REL}}^{(l)} \left[\mathbf{m}_{\xi \rightarrow (p,e)}^{(l)}; \mathbf{e}^{(l-1)} \right]$



04 Contextual Message Passing

- Updates entity and relation representations by **aggregating messages from the facts they belong to** while explicitly **excluding their own information** from those facts' messages
 - Forces the representations to rely solely on the surrounding **context** in the facts
- To update a component within a fact, we compute a **context message** that encodes **all the other components in the fact**
 - The context messages from fact ξ to its components e and p in the constituent air (p, e) are computed by:

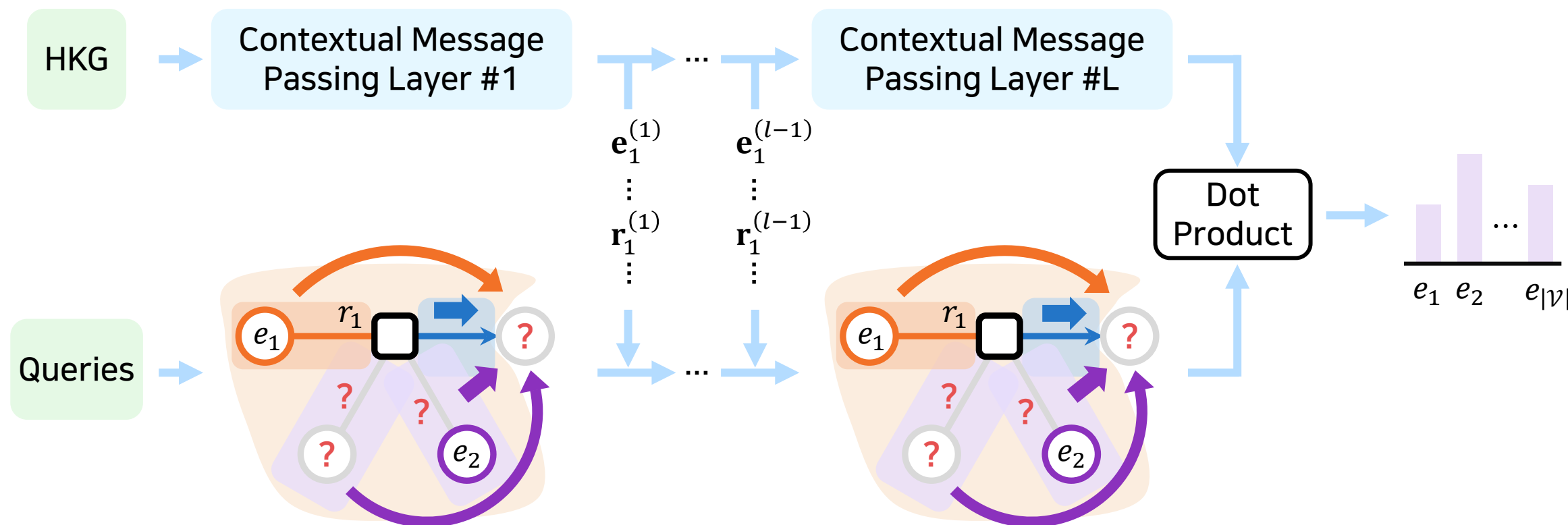
- $\mathbf{m}_{\xi \rightarrow (p,e)}^{(l)} = \text{MLP}^{(l)} \left(\frac{\mathbf{z}_{\xi}^{(l)} - \mathbf{z}_{(p,e)}^{(l)}}{n_{\xi} + 1} \right)$
- $\mathbf{m}_{\xi \rightarrow e}^{(l)} = \mathbf{W}_{\text{ENT}}^{(l)} \left[\mathbf{m}_{\xi \rightarrow (p,e)}^{(l)}; \mathbf{p}^{(l-1)} \right]$
- $\mathbf{m}_{\xi \rightarrow p}^{(l)} = \mathbf{W}_{\text{REL}}^{(l)} \left[\mathbf{m}_{\xi \rightarrow (p,e)}^{(l)}; \mathbf{e}^{(l-1)} \right]$



04

Probabilistic Masked Fact Decoding

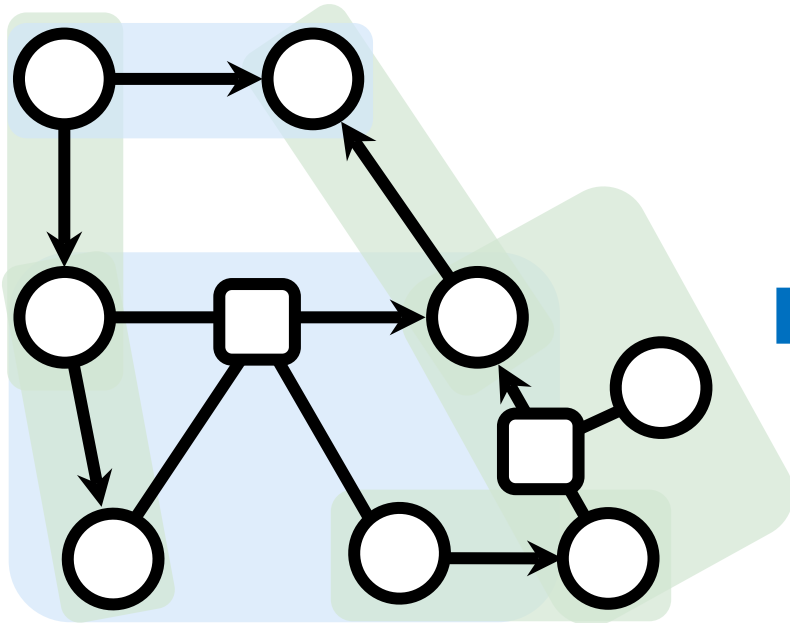
- Treats each masked entity and relation as a vanilla entity and relation, respectively, and updates their representations via contextual message passing
 - A masked fact ζ is used **only to update its masked components**



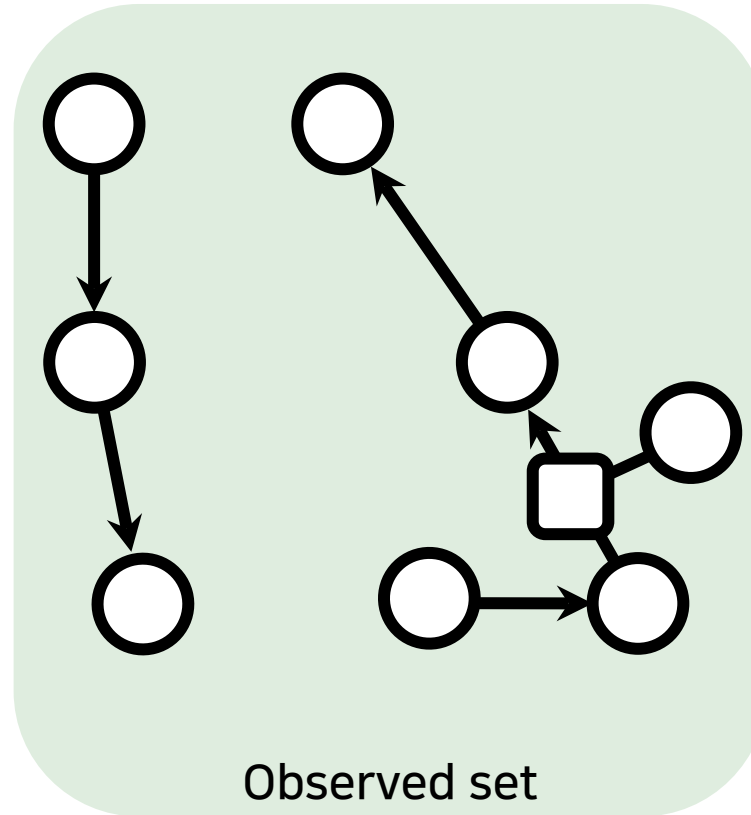
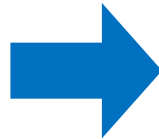
04

Bi-level Noising Strategy

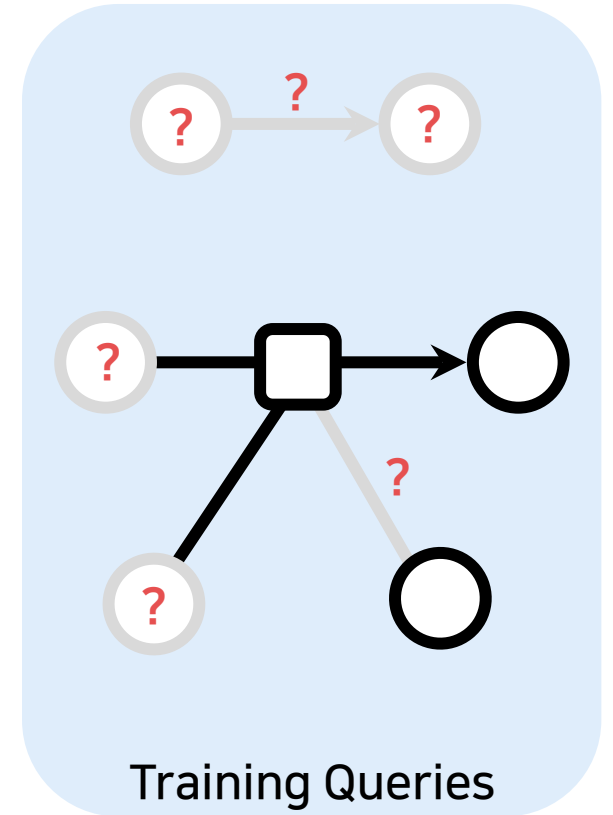
- Simulates **diverse conditional generation scenarios** by varying both the observed structure and the query during training



Training Set



Observed set



Training Queries

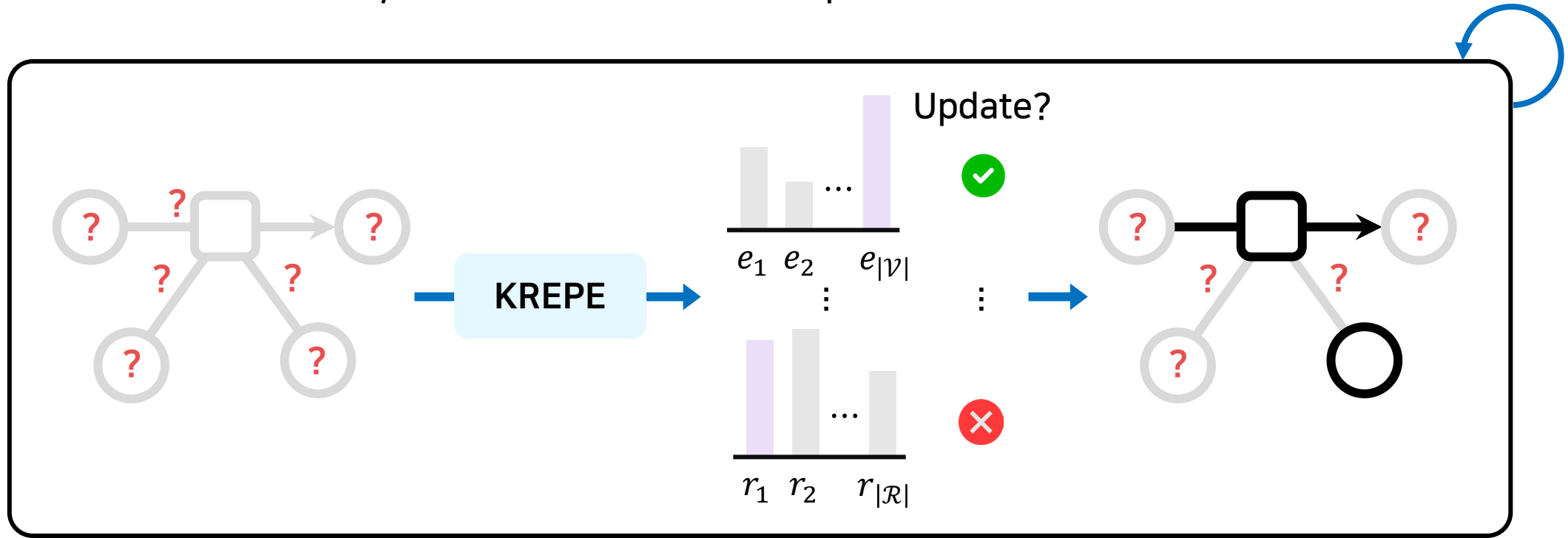
04 Discriminative and Generative Inference

- Link prediction: leverages the computed probability distribution to rank the candidates



04 Discriminative and Generative Inference

- Fact Generation: Iteratively decodes the masked components



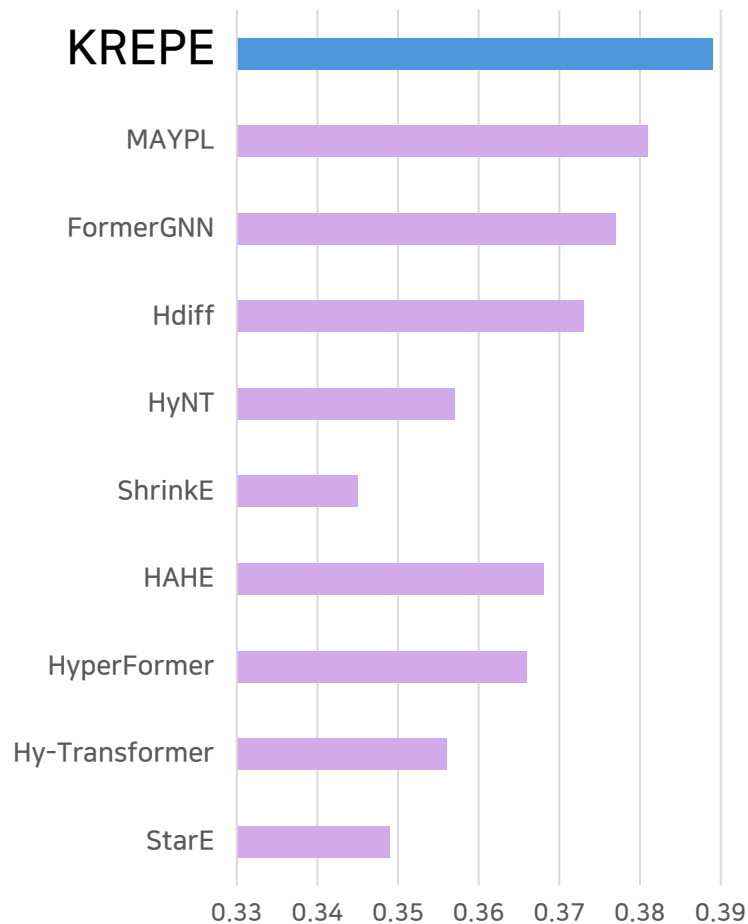
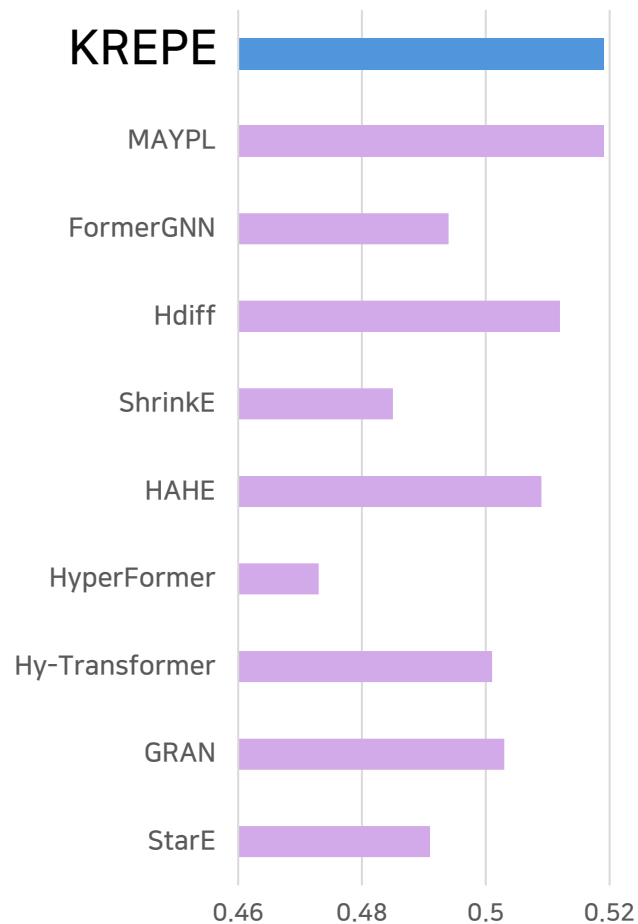
((Gigi, awarded, Oscars Best Adapted Screenplay), {(subject of, 31st Oscars), (winner, A. J. Lerner)})

04 Experimental Results

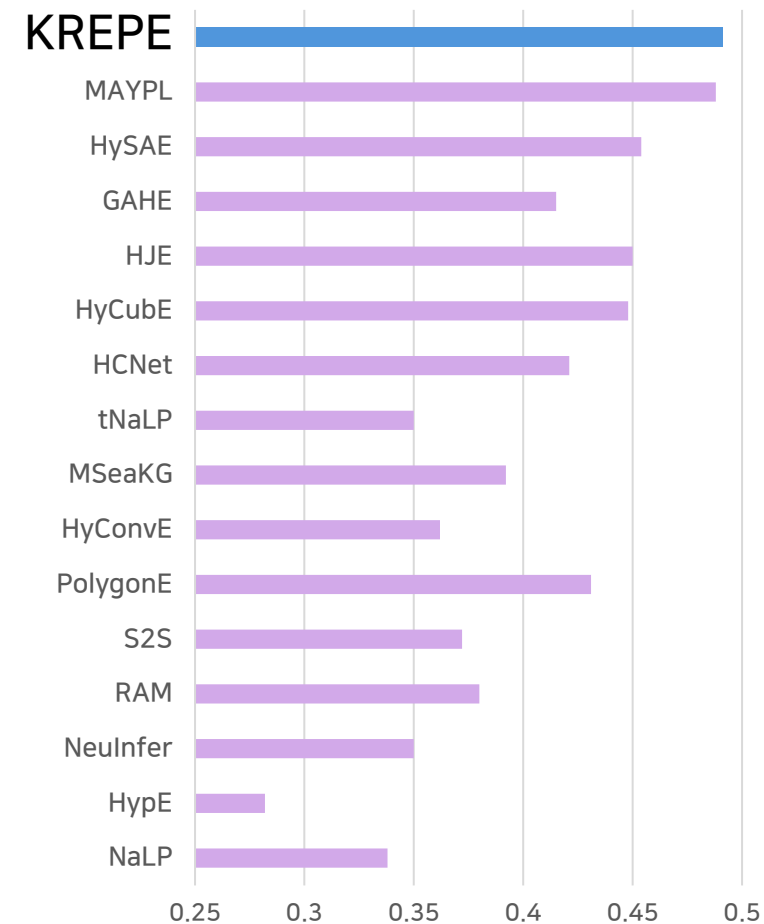
- Datasets
 - 3 HKG datasets: WD50K, WikiPeople⁻, WikiPeople
- Baselines
 - 25 link prediction methods and seven competitive fact generation baselines
 - Fact generation baselines include two variations of discriminative HKG models and five LLM-based approaches
- Fact Generation Settings
 - Scratch: generating a valid fact without any known component
 - Targeted: requiring the generation of a fact containing a single given component called a target
 - Arbitrary Masking: generating a fact from an arbitrarily masked input

Link Prediction Results

WD50K

WikiPeople⁻

WikiPeople



04 Fact Generation Baselines

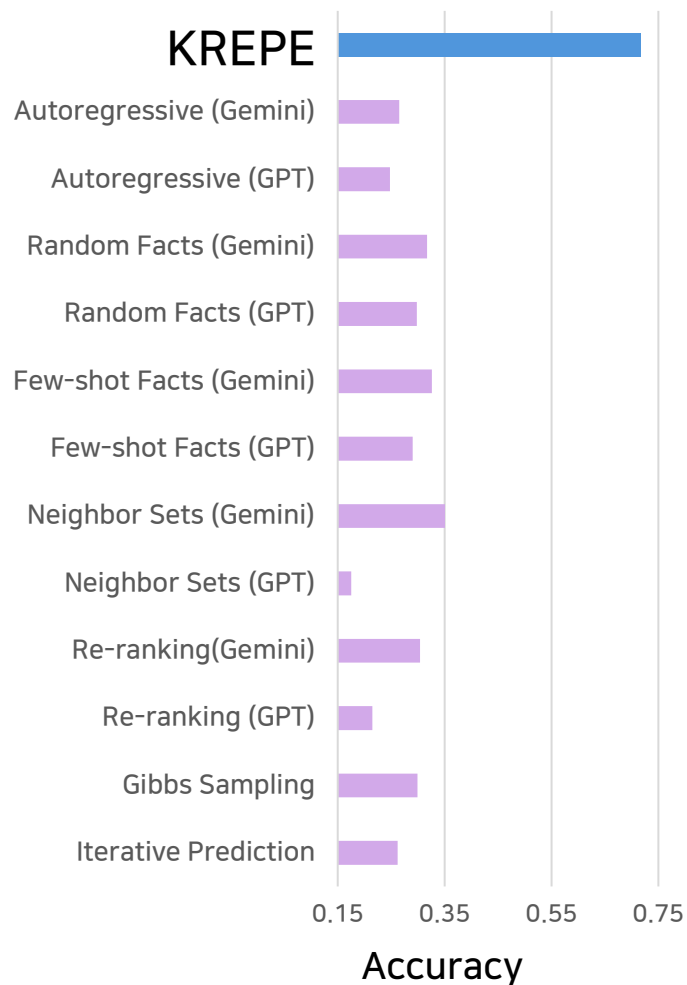
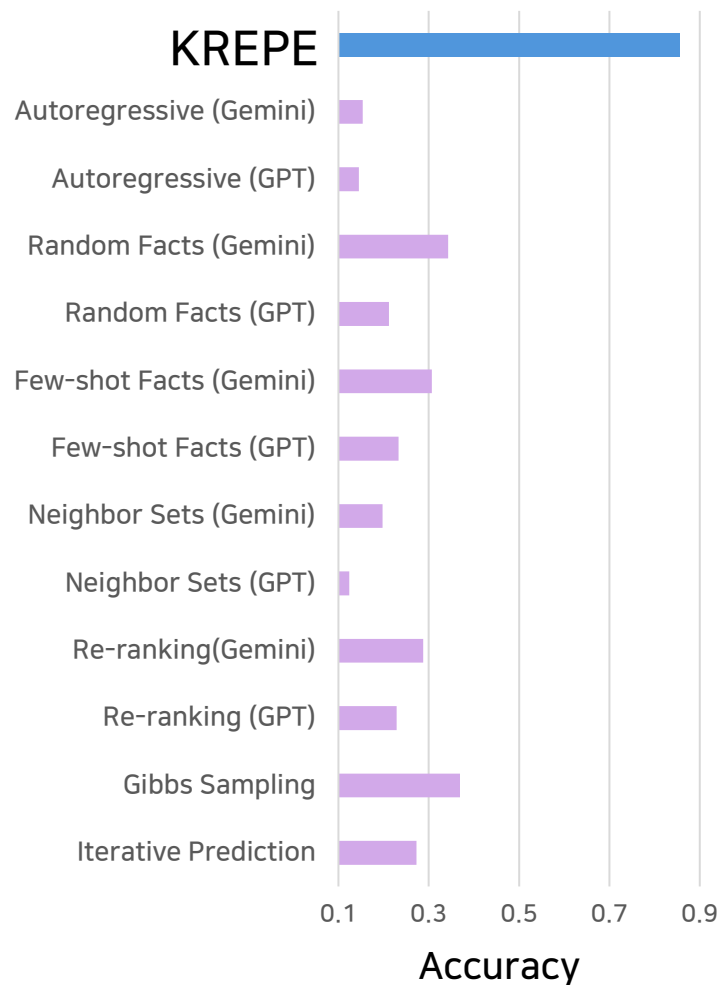
- Iterative Prediction (based on an HKG method)
 - Missing components are randomly initialized, and **sequentially masked and replaced by an HKG model**
- Gibbs Sampling (based on an HKG method)
 - Missing components are randomly initialized, and **iteratively masked and replaced by resampling from the distribution predicted by an HKG model**
- Re-ranking (based on an HKG method and an LLM)
 - Missing components are initialized using a training fact that matches the query. Each missing component is sequentially masked, and **an LLM selects the best candidate from the top-5 predictions of an HKG model**

- Neighbor Sets (based on an LLM)
 - An LLM generates each fact prompted by the **neighbors of known components in the query**
- Few-shot Facts (based on an LLM)
 - An LLM generates each fact prompted by **30 training facts containing at least one known component in the query**
- Random Facts (based on an LLM)
 - An LLM is prompted to generate facts by utilizing **1,000 randomly sampled training facts**
- Autoregressive (based on an LLM)
 - An LLM generates components **step-by-step**, prompted to utilize **training facts containing the previously predicted component for generation**

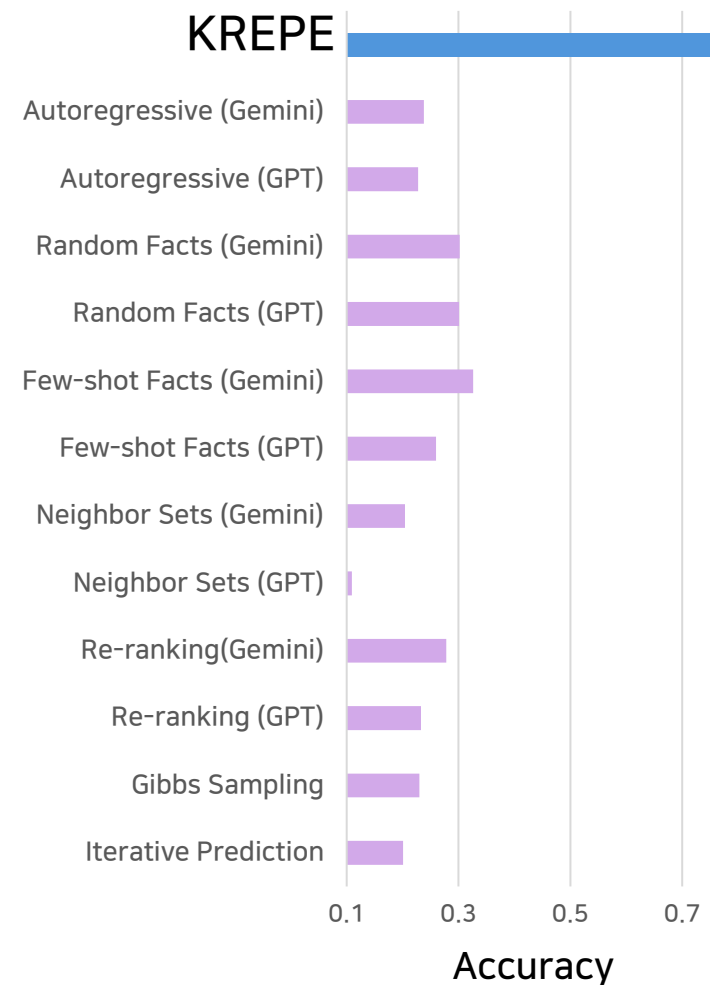
04

Fact Generation Results - Scratch

WD50K

WikiPeople⁻

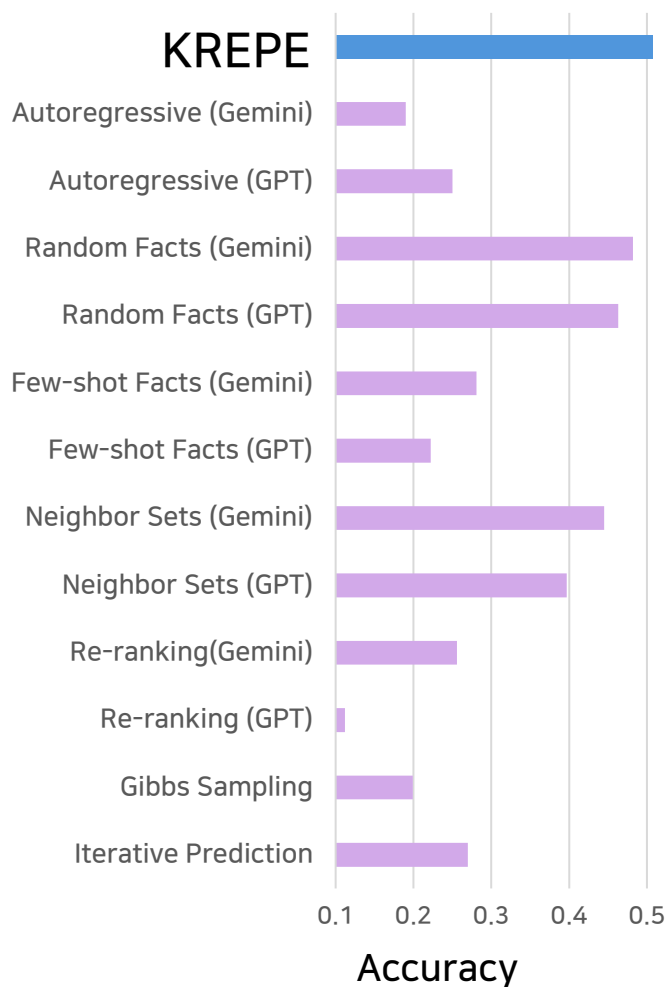
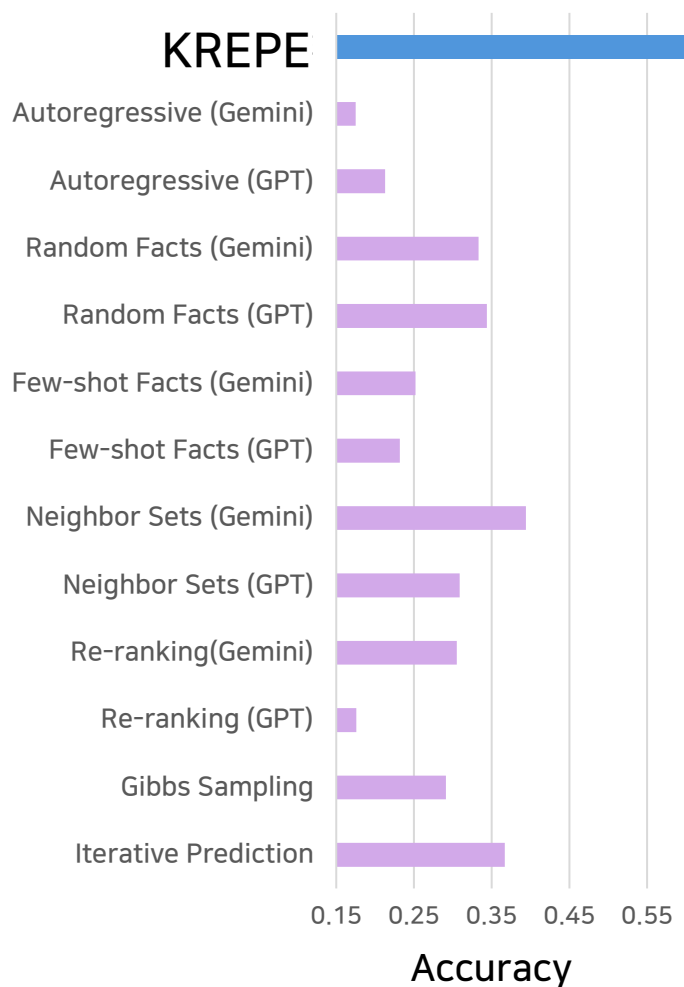
WikiPeople



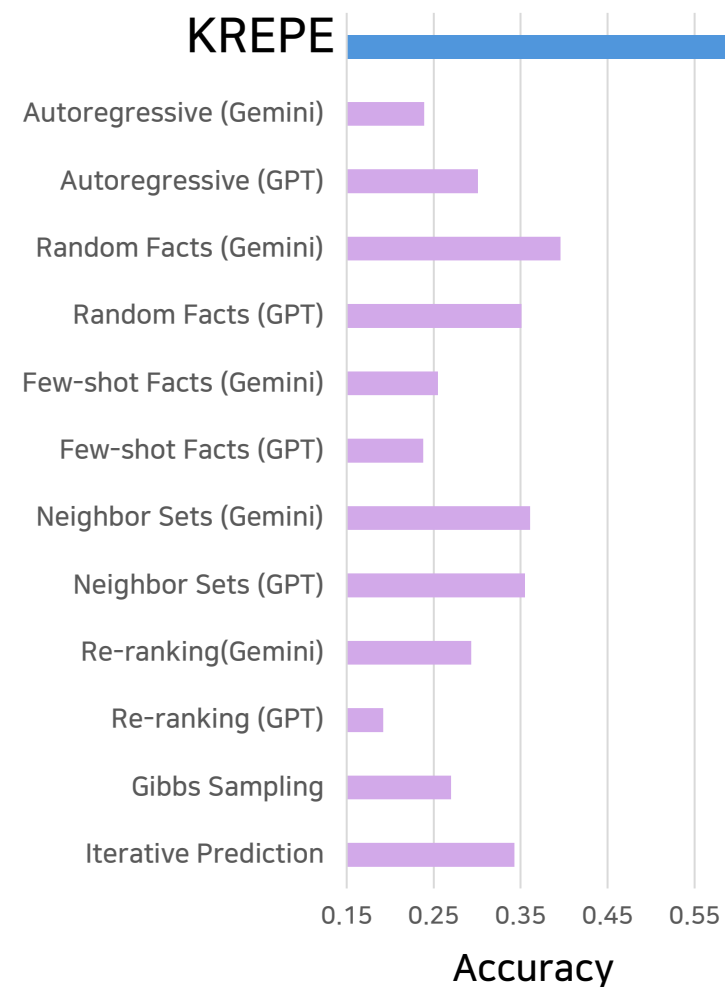
04

Fact Generation Results - Targeted

WD50K

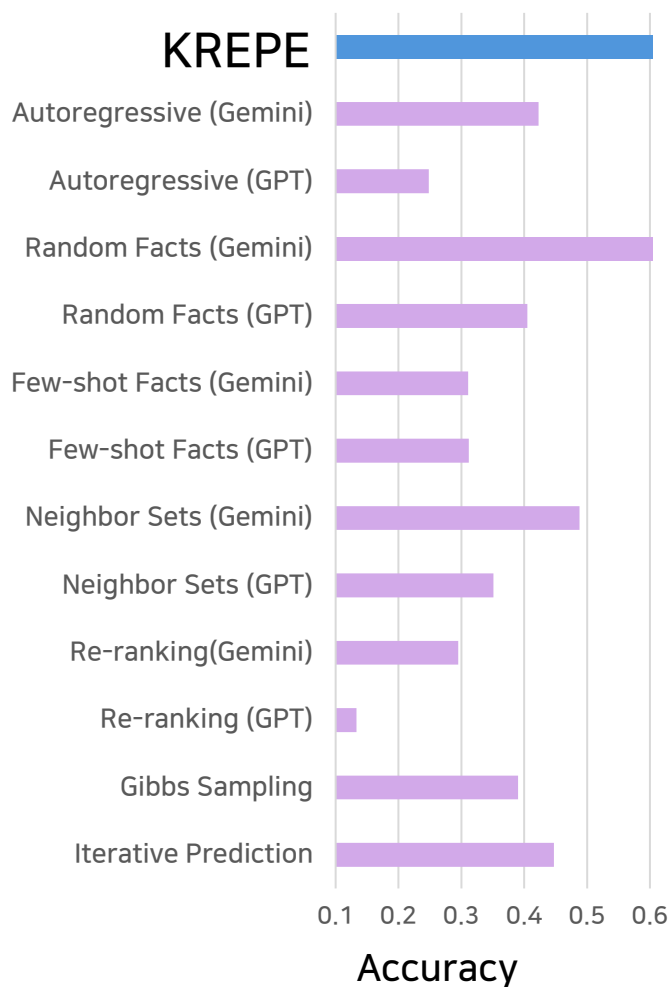
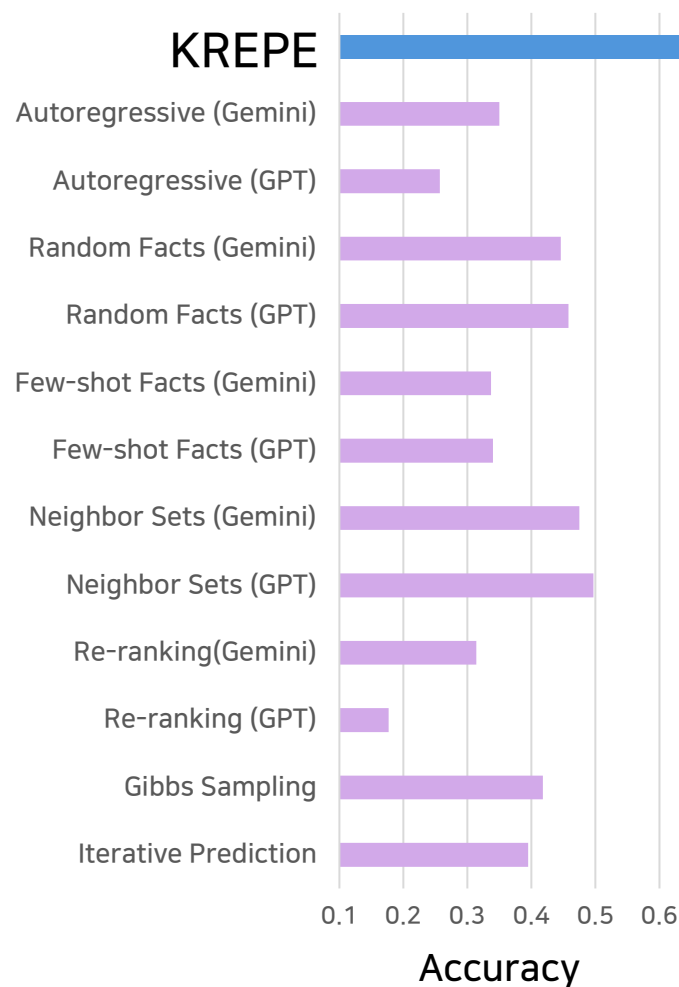
WikiPeople⁻

WikiPeople

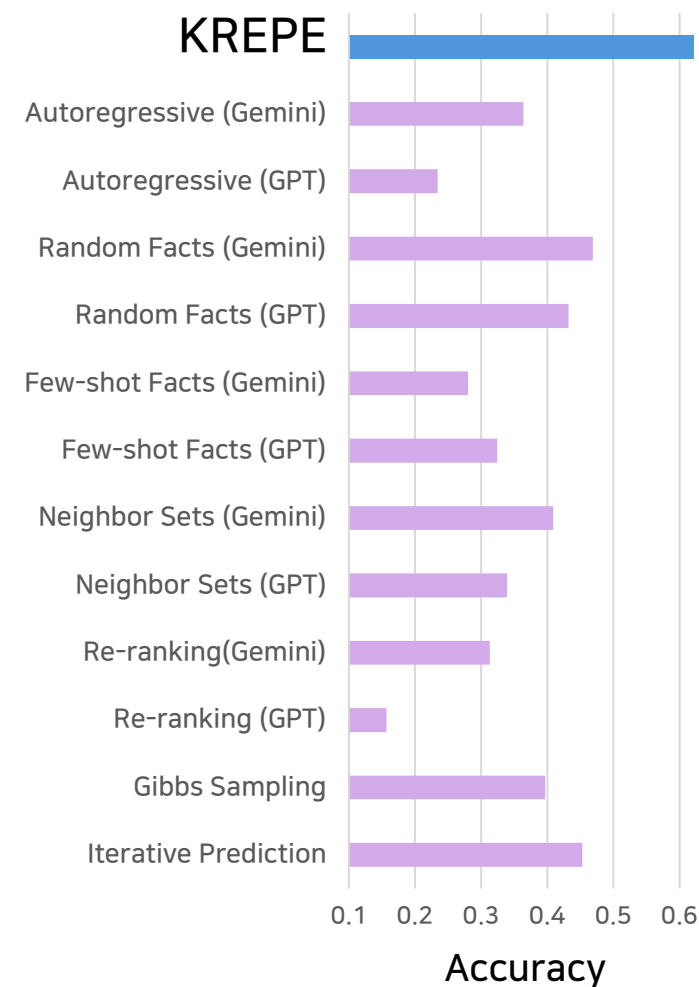


Fact Generation Results - Arbitrary

WD50K

WikiPeople⁻

WikiPeople



04 Conclusion

- Introduce **Fact Generation**, a novel task that aligns closely with real-world scenarios where **multiple components of a fact may be missing**
 - Advances beyond the traditional fill-in-the-blank link prediction in the HKG domain
- Propose **KREPE**, the first **HKG representation learning framework equipped with generative capabilities** via a masked discrete diffusion objective
- KREPE not only establishes a new state-of-the-art link prediction performance but also significantly outperforms LLM-based baselines in fact generation

05 References

- Some slides are made based on the following references.
 - P. Rosso et al., “Beyond Triplets: Hyper-Relational Knowledge Graph Embedding for Link Prediction”, TheWebConf, 2020.
 - M. Galkin et al., “Message Passing for Hyper-Relational Knowledge Graphs”, EMNLP, 2020.
 - Q. Wang et al., “Link Prediction on N-ary Relational Facts: A Graph-based Approach”, ACL Findings, 2020.
 - H. Luo et al., “HAHE: Hierarchical Attention for Hyper-Relational Knowledge Graphs in Global and Local Level”, ACL, 2023.
 - B. Xiong et al., “Shrinking Embeddings for Hyper-Relational Knowledge Graphs”, ACL, 2023.
 - C. Chung et al., “Representation Learning on Hyper-Relational and Numeric Knowledge Graphs with Transformers”, KDD, 2023.
 - Z. Hu et al., “HyperFormer: Enhancing Entity and Relation Interaction for Hyper-Relational Knowledge Graph Completion”, CIKM, 2023.
 - J. Lee and J. J. Whang, “Structure Is All You Need: Structural Representation Learning on Hyper-Relational Knowledge Graphs”, ICML, 2025.
 - J. Lee, S. Kim, and J. J. Whang, “Generative Representation Learning on Hyper-relational Knowledge Graphs via Masked Discrete Diffusion”, ICML 2026.